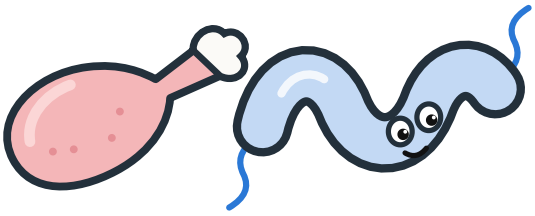
MICRO-ORGANISMS 01

Campylobacter

Bacterium



WHAT IT DOES

Diarrhoea and fever; one in four ends up in hospital

WHERE IT IS FOUND

Poultry meat (especially chicken), unpasteurised milk, unwashed vegetables, contaminated water

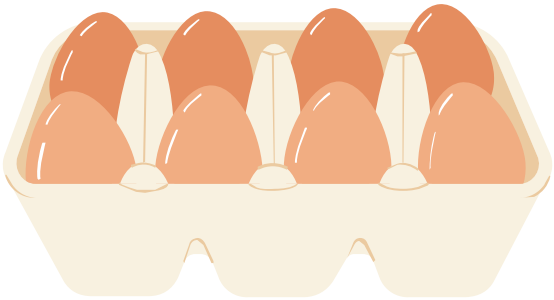
HOW OFTEN IT IS FOUND: In Latvia, about a third of fresh chicken samples carry Campylobacter

HOW OFTEN IT IS EATEN: Chicken gets cooked, and heat kills them; live ones only now and then

MICRO-ORGANISMS 02

Salmonella

Bacterium



WHAT IT DOES

Diarrhoea and fever; often needs hospital care

WHERE IT IS FOUND

Eggs and egg products, chicken and pork, undercooked dishes

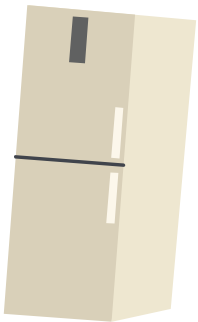
HOW OFTEN IT IS FOUND: Rarely found in Latvian eggs and poultry, thanks to control programmes

HOW OFTEN IT IS EATEN: Eggs and meat, almost daily

MICRO-ORGANISMS 03

Listeria monocytogenes

Bacterium



WHAT IT DOES

Dangerous in pregnancy and old age; can be fatal

WHERE IT IS FOUND

Cold-smoked fish (salmon), ready-to-eat meals, soft cheese, deli meat, bagged salad

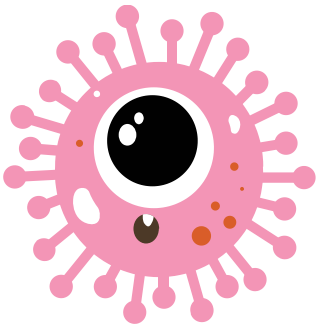
HOW OFTEN IT IS FOUND: Grows even in the fridge: there is more of it every day

HOW OFTEN IT IS EATEN: Ready meals and cheese, often

MICRO-ORGANISMS 04

Norovirus

Virus



WHAT IT DOES

Vomiting and diarrhoea for a few days

WHERE IT IS FOUND

Prepared dishes, frozen berries, shellfish, ice, person to person

HOW OFTEN IT IS FOUND: Frozen berries: over 50 norovirus alerts in Europe since 2014

HOW OFTEN IT IS EATEN: Eaten daily, including away from home

RISK ASSESSORS PUT IT HERE

SERIOUS × RARELY

severity × exposure





MICRO-ORGANISMS

Salmonella

02

Second most common infection passed from animals to people in the European Union (EU); eggs are the leading source. Latvia's notification rate is among the lowest in the EU and has fallen sharply, so here it sits under “rarely”.

HOW MUCH IS TOO MUCH?

No safe dose; what counts is how many multiplied while the food stood warm.

Source: European Food Safety Authority (EFSA) and European Centre for Disease Prevention and Control (ECDC), 2024; ECDC, Salmonellosis: Annual Epidemiological Report 2023

RISK ASSESSORS PUT IT HERE

SERIOUS × SOMETIMES

severity × exposure





MICRO-ORGANISMS

Campylobacter

01

The most reported foodborne illness in Europe. Latvia reports about 5× fewer cases than the European Union average. The European Centre for Disease Prevention and Control (ECDC) attributes this to under-detection, not to safer chicken.

HOW MUCH IS TOO MUCH?

There is no safe dose: a few hundred bacteria are enough. Only cooking matters.

Source: European Food Safety Authority (EFSA) and European Centre for Disease Prevention and Control (ECDC), One Health Zoonoses Report 2024; ECDC, Annual Epidemiological Report

RISK ASSESSORS PUT IT HERE

MILD × OFTEN

severity × exposure





MICRO-ORGANISMS

Norovirus

04

The biggest cause of outbreaks by number of people affected, yet almost never fatal in healthy adults. The textbook “often, mild consequences” case.

HOW MUCH IS TOO MUCH?

No safe dose; a few dozen virus particles can infect.

Source: European Food Safety Authority (EFSA) and European Centre for Disease Prevention and Control (ECDC), One Health Zoonoses Report 2024

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





MICRO-ORGANISMS

Listeria monocytogenes

03

About 55 times rarer than Campylobacter infection, yet kills more people. Almost every case is hospitalised. Unlike most bacteria it grows at refrigerator temperature.

HOW MUCH IS TOO MUCH?

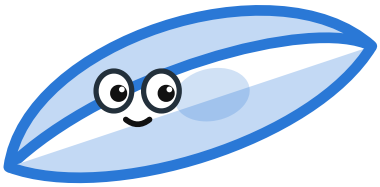
The limit in ready-to-eat food is 100 bacteria per gram: no concern for a healthy adult, but a concern in pregnancy.

Source: European Food Safety Authority (EFSA) and European Centre for Disease Prevention and Control (ECDC), One Health Zoonoses Report 2024

MICRO-ORGANISMS 05

Toxoplasma gondii

Parasite



↑ WHAT IT DOES

Usually symptomless; can harm the fetus in pregnancy

→ WHERE IT IS FOUND

Undercooked meat, cured meat, unwashed vegetables, cat faeces

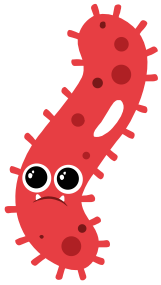
HOW OFTEN IT IS FOUND: Many people take it in during their life, usually without noticing

HOW OFTEN IT IS EATEN: Meat often; unwashed vegetables sometimes

MICRO-ORGANISMS 06

Shiga toxin-producing E. coli

Bacterium: Shiga toxin-producing Escherichia coli (STEC)



↑ WHAT IT DOES

Can cause kidney failure, especially in children

→ WHERE IT IS FOUND

Minced beef, raw milk, unwashed vegetables, sprouts, flour, contaminated water

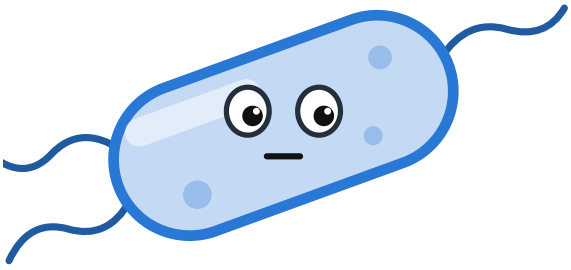
HOW OFTEN IT IS FOUND: Even in flour: live STEC in one in ten ready-made doughs in Germany

HOW OFTEN IT IS EATEN: Minced meat, several times a month

MICRO-ORGANISMS 07

Yersinia enterocolitica

Bacterium



↑ WHAT IT DOES

Abdominal pain; can mimic appendicitis

→ WHERE IT IS FOUND

Pork, especially freshly minced; unpasteurised milk

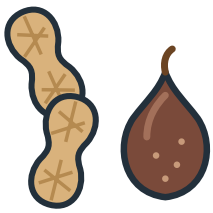
HOW OFTEN IT IS FOUND: The fourth most common infection passed from animals to people in the European Union, yet few cases are registered

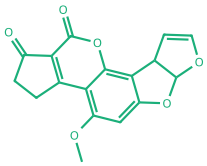
HOW OFTEN IT IS EATEN: Pork is eaten often

MYCOTOXINS 08

Aflatoxins

$C_{17}H_{12}O_6$ aflatoxin B1





↑ WHAT IT DOES

Damages DNA; causes liver cancer in humans

→ WHERE IT IS FOUND

Nuts (peanuts, pistachios, almonds), dried figs, maize, dried spices, milk (aflatoxin M1)

HOW OFTEN IT IS FOUND: European Union checks rarely find levels over the limit in imported shipments

HOW OFTEN IT IS EATEN: Nuts and figs, now and then

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





MICRO-ORGANISMS

06

Shiga toxin-producing E. coli

Rare but dangerous, especially for young children. In 2011 Germany publicly blamed Spanish cucumbers; the real source was sprouts. Spanish growers lost about $200 million a week while wrongly accused.

HOW MUCH IS TOO MUCH?

No safe dose; a handful of bacteria is enough, so cooking decides, not quantity.

Source: European Food Safety Authority (EFSA) and European Centre for Disease Prevention and Control (ECDC), 2024; Food Safety News, 2015

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × SOMETIMES

severity × exposure





MICRO-ORGANISMS

05

Toxoplasma gondii

Ranked first of 20 pathogens by the Norwegian Scientific Committee for Food and Environment (VKM) in 2021, above Campylobacter and Salmonella. Usually symptomless, but consequences in pregnancy can be permanent.

HOW MUCH IS TOO MUCH?

No safe dose; what counts is whether the meat was hot enough or frozen.

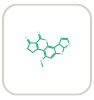
Source: Norwegian Scientific Committee for Food and Environment (VKM), Report 2021:10

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





MYCOTOXINS

08

Aflatoxins

The most potent natural carcinogen in food; no safe level is set. Yet European exposure is low: aflatoxins are among the most frequent rapid alerts in the European Union, and most contaminated shipments are stopped at the border. High hazard, controlled exposure.

HOW MUCH IS TOO MUCH?

Aflatoxins damage DNA, so no safe dose is set. The limit is as low as is technically achievable.

Source: European Food Safety Authority (EFSA), 2020; Ruokavirasto 1/2021; Rapid Alert System for Food and Feed (RASFF)

RISK ASSESSORS PUT IT HERE

SERIOUS × RARELY

severity × exposure





MICRO-ORGANISMS

07

Yersinia enterocolitica

Fourth most common infection passed from animals to people in the European Union, mainly from pork. Can mimic appendicitis: another example of reported numbers depending on how hard you look.

HOW MUCH IS TOO MUCH?

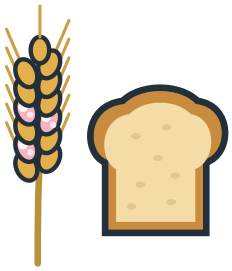
No safe dose; the cooking step decides.

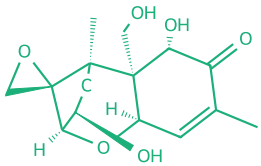
Source: European Food Safety Authority (EFSA) and European Centre for Disease Prevention and Control (ECDC), One Health Zoonoses Report

MYCOTOXINS 09

Deoxynivalenol (DON)

$C_{15}H_{20}O_6$ "vomitoxin"





WHAT IT DOES

Nausea, vomiting, loss of appetite

WHERE IT IS FOUND

Bread, porridge, wheat and barley products, breakfast cereals, pasta, beer

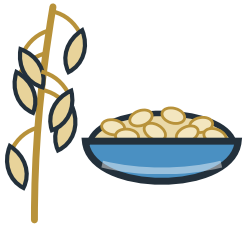
HOW OFTEN IT IS FOUND: The most commonly found mycotoxin in European grain

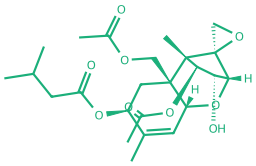
HOW OFTEN IT IS EATEN: Bread and porridge, every day

MYCOTOXINS 10

T-2 and HT-2 toxins

$C_{24}H_{34}O_9$ T-2 toxin





WHAT IT DOES

Weakens the immune system; irritates the gut

WHERE IT IS FOUND

Oats and oat flakes (also muesli), barley, rye, oat drinks

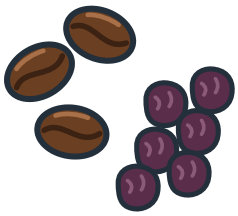
HOW OFTEN IT IS FOUND: Mostly in oats, and found more often in a cool, damp climate

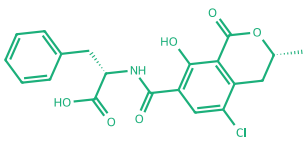
HOW OFTEN IT IS EATEN: Oat flakes, every morning for many

MYCOTOXINS 11

Ochratoxin A

$C_{20}H_{18}ClNO_6$ ochratoxin A





WHAT IT DOES

Damages the kidneys; considered possibly carcinogenic

WHERE IT IS FOUND

Cereals, coffee (roasted and instant), raisins, wine, dried spices (paprika, chilli), cured meat

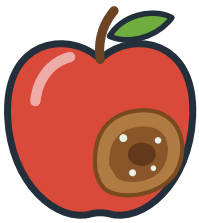
HOW OFTEN IT IS FOUND: In many everyday products, but usually in small amounts

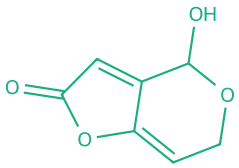
HOW OFTEN IT IS EATEN: Coffee and cereals, every day

MYCOTOXINS 12

Patulin

$C_7H_6O_4$ patulin





WHAT IT DOES

Irritates the gut at high doses

WHERE IT IS FOUND

Apple juice, apple purée, cider, rotten apples and pears

HOW OFTEN IT IS FOUND: Forms only in rotten fruit, which is sorted out before pressing

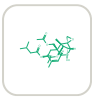
HOW OFTEN IT IS EATEN: Apple juice often, rotten apples rarely

RISK ASSESSORS PUT IT HERE

SERIOUS × SOMETIMES

severity × exposure





MYCOTOXINS

T-2 and HT-2 toxins

10

Scored joint-highest of all chemicals for monitoring priority by the Norwegian Scientific Committee for Food and Environment (VKM) in 2019, above lead, cadmium and mercury. Oats are the most affected cereal. Binding European Union limits only came into force in 2024.

HOW MUCH IS TOO MUCH?

The limit is based on immune effects; exceeding it means a smaller margin, not illness.

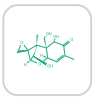
Source: Norwegian Scientific Committee for Food and Environment (VKM), 2019:13; Regulation (EU) 2024/1038

RISK ASSESSORS PUT IT HERE

MILD × OFTEN

severity × exposure





MYCOTOXINS

Deoxynivalenol (DON)

09

The mycotoxin that actually exceeds its tolerable intake in Europe (in infants, toddlers and children). Not exotic food: ordinary bread and porridge. Effects are usually mild and reversible.

HOW MUCH IS TOO MUCH?

There are two limits: a daily one protecting children's growth, and a single-occasion one eight times higher protecting against vomiting.

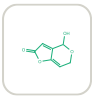
Source: Regulation (EU) 2024/1022, recitals; European Environment Agency (EEA), 2025

RISK ASSESSORS PUT IT HERE

MILD × RARELY

severity × exposure





MYCOTOXINS

Patulin

12

Forms when apples go mouldy, which is why juice is pressed from sound fruit. The European Union limit for infant apple products is five times stricter than for ordinary juice.

HOW MUCH IS TOO MUCH?

The limit is based on irritation; reaching it takes a lot of juice from rotten apples.

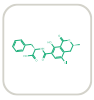
Source: Regulation (EU) 2023/915; Ruokavirasto 1/2021

RISK ASSESSORS PUT IT HERE

SERIOUS × SOMETIMES

severity × exposure





MYCOTOXINS

Ochratoxin A

11

Affects the kidneys. The European Food Safety Authority (EFSA) set a tolerable weekly intake in 2006 and then WITHDREW it in 2020, concluding a safe level could not be justified. The old number still circulates.

HOW MUCH IS TOO MUCH?

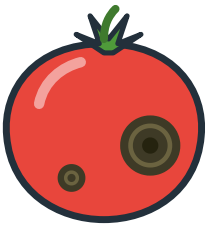
The limit protects the kidneys, and harm comes from years of intake, not one loaf.

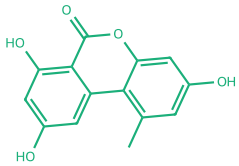
Source: European Food Safety Authority (EFSA), 2020 opinion on ochratoxin A

MYCOTOXINS 13

Alternaria toxins

$C_{14}H_{10}O_5$ alternariol (AOH)





↑ WHAT IT DOES

Damages DNA; may cause cancer

→ WHERE IT IS FOUND

Tomatoes and tomato products, sunflower seeds, cereals, fruit juices

HOW OFTEN IT IS FOUND: A very common mould; its toxins turn up in many plant products

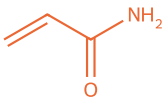
HOW OFTEN IT IS EATEN: Tomatoes and juice, almost daily

PROCESS CONTAMINANTS 14

Acrylamide

C_3H_5NO acrylamide





↑ WHAT IT DOES

Damages DNA; considered probably carcinogenic

→ WHERE IT IS FOUND

Fries, crisps, toast, roasted coffee, biscuits and gingerbread, breakfast cereals

HOW OFTEN IT IS FOUND: Forms whenever starchy food is fried or roasted above 120 °C

HOW OFTEN IT IS EATEN: Fried and roasted food, every day

PROCESS CONTAMINANTS 15

Furan

C_4H_4O furan





↑ WHAT IT DOES

Damages the liver; considered possibly carcinogenic

→ WHERE IT IS FOUND

Coffee (roasted and instant), jarred baby food, canned food, soups

HOW OFTEN IT IS FOUND: Forms during heat treatment; most of it in coffee and canned food

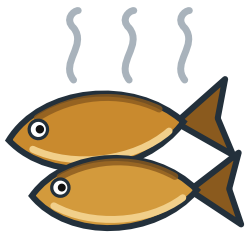
HOW OFTEN IT IS EATEN: Coffee, every day

PROCESS CONTAMINANTS 16

Benzo(a)pyrene and PAHs

$C_{20}H_{12}$

benzo[a]pyrene and polycyclic aromatic hydrocarbons (PAHs)





↑ WHAT IT DOES

Damages DNA at high doses

→ WHERE IT IS FOUND

Smoked fish (sprats, herring, salmon), smoked meat, grilled and chargrilled food

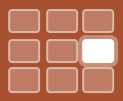
HOW OFTEN IT IS FOUND: In one study, some samples of Latvian smoked meat were over the limit

HOW OFTEN IT IS EATEN: Smoked and grilled, a few times a month

RISK ASSESSORS PUT IT HERE

SERIOUS × OFTEN

severity × exposure





PROCESS CONTAMINANTS

14

Acrylamide

Forms when starch is fried or roasted above 120 °C: not added, created by cooking. In one Canadian study 34% of the public ranked it the top food risk; 0% of experts did.

HOW MUCH IS TOO MUCH?

There is no limit at all. What is measured is the gap to the dose that caused tumours in animals: currently a few hundred times, not the ten thousand times that would mean low concern.

Source: European Food Safety Authority (EFSA), 2015; Ruokavirasto 1/2021; Webster and colleagues, 2010

RISK ASSESSORS PUT IT HERE

SERIOUS × OFTEN

severity × exposure





MYCOTOXINS

13

Alternaria toxins

Placed in the HIGHEST risk group by Finland's 2021 assessment, alongside aflatoxins, inorganic arsenic and acrylamide. Almost nobody has heard of them, and the European Union has no maximum levels for them, only indicative levels for monitoring since 2022.

HOW MUCH IS TOO MUCH?

There is no maximum level. Since 2022 there are indicative levels for monitoring, for example 500 µg/kg of tenuazonic acid in processed tomato products; they are not food safety limits.

Source: Ruokavirasto 1/2021, Finland; European Food Safety Authority (EFSA), dietary exposure to Alternaria toxins, 2016; Commission Recommendation (EU) 2022/553 on monitoring Alternaria toxins in food

RISK ASSESSORS PUT IT HERE

SERIOUS × SOMETIMES

severity × exposure





PROCESS CONTAMINANTS

16

Benzo(a)pyrene and PAHs

A Latvian study of smoked meat found some samples over the legal limit, yet calculated exposure stayed far from the level of concern. The European Union sets a special higher limit for smoked sprats and small Baltic herring. Over the limit is not the same as harmful.

HOW MUCH IS TOO MUCH?

Smoked fish does have a limit. Over the limit means the process went wrong, not that the food is poisonous.

Source: Food Control, 2015 (Latvia); Regulation (EU) 2023/915; Ruokavirasto 1/2021

RISK ASSESSORS PUT IT HERE

SERIOUS × SOMETIMES

severity × exposure





PROCESS CONTAMINANTS

15

Furan

Near the top in both the Finnish and Norwegian assessments, and essentially unknown to the public. It is volatile: most of it leaves during brewing. What is in the product and what reaches the body are not the same thing.

HOW MUCH IS TOO MUCH?

No limit; what is measured is the gap to the dose that damaged animals' livers, and it is small.

Source: European Food Safety Authority (EFSA), 2017; Ruokavirasto 1/2021; Norwegian Scientific Committee for Food and Environment (VKM), 2019:13

PROCESS CONTAMINANTS 17

Glycidyl esters

$C_3H_6O_2$ glycidol





WHAT IT DOES

Turns into glycidol in the body, which damages DNA; probably carcinogenic

WHERE IT IS FOUND

Refined vegetable oils, palm oil, margarine, infant formula, confectionery

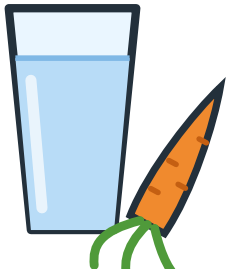
HOW OFTEN IT IS FOUND: Forms whenever oil is refined at high temperature

HOW OFTEN IT IS EATEN: Processed food, every day

METALS 18

Lead

plumbum





WHAT IT DOES

Even at low doses it slightly affects children's brain development

WHERE IT IS FOUND

Cereals, vegetables, beverages, game meat, supplements, drinking water

HOW OFTEN IT IS FOUND: In small amounts in almost all food and water

HOW OFTEN IT IS EATEN: Everyday food and water, always

METALS 19

Cadmium

cadmium





WHAT IT DOES

Builds up in the kidneys over decades; at high doses weakens the bones

WHERE IT IS FOUND

Cereals, vegetables (potatoes, root vegetables), mushrooms, cocoa and chocolate, molluscs, tobacco

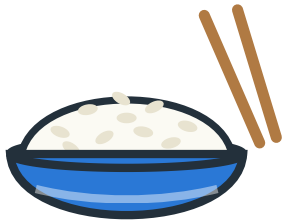
HOW OFTEN IT IS FOUND: In almost all everyday plant food

HOW OFTEN IT IS EATEN: Cereals and vegetables, every day

METALS 20

Inorganic arsenic

arsenicum





WHAT IT DOES

Damages DNA; causes skin and lung cancer in humans

WHERE IT IS FOUND

Rice and rice products (rice flakes, rice cakes, rice drink), drinking water, fruit juice

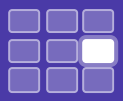
HOW OFTEN IT IS FOUND: Mainly in rice; less in other food

HOW OFTEN IT IS EATEN: Rice, a few times a week

RISK ASSESSORS PUT IT HERE

SERIOUS × OFTEN

severity × exposure



82 Pb

METALS

Lead

18

No safe level is set; it affects neurodevelopment in children. It ranks first among five toxic elements in our own Latvian data, yet two in three of 122,572 measurements found nothing at all.

HOW MUCH IS TOO MUCH?

No tolerable intake: there is no safe level for children's development. The goal is not “under the limit” but “as low as possible”.

Source: Institute of Food Safety, Animal Health and Environment “BIOR” / Data-to-Decisions, Latvia × adults; European Food Safety Authority (EFSA)

RISK ASSESSORS PUT IT HERE

SERIOUS × SOMETIMES

severity × exposure



PROCESS CONTAMINANTS

Glycidyl esters

17

Formed during high-temperature refining of vegetable oils. The infant-formula limit is roughly 20–150× stricter than for ordinary oil: same hazard, different limits, because who is exposed and how much differs.

HOW MUCH IS TOO MUCH?

The limit is set for infants, because formula can be their only food.

Source: Regulation (EU) 2023/915; Norwegian Scientific Committee for Food and Environment (VKM), 2019:13

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × SOMETIMES

severity × exposure



33 As

METALS

Inorganic arsenic

20

The top foodborne chemical globally in the 2026 burden estimate of the World Health Organization (WHO). European exposure is far lower than in South Asia, where it enters groundwater. Same substance, very different risk, because exposure differs.

HOW MUCH IS TOO MUCH?

No tolerable intake; what is measured is the gap to the dose at which skin lesions and cancer were seen in people.

Source: World Health Organization (WHO), 2026, foodborne disease burden; Ruokavirasto 1/2021

RISK ASSESSORS PUT IT HERE

MILD × OFTEN

severity × exposure



48 Cd

METALS

Cadmium

19

Accumulates in the kidneys over decades; everyone is exposed. Average Latvian intake sits slightly ABOVE the guidance value, which does not mean harm. The guidance value has a large safety margin built in; it means less margin. That is why we monitor it.

HOW MUCH IS TOO MUCH?

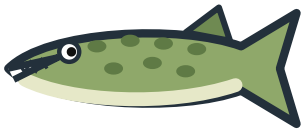
The limit is weekly, not daily: cadmium stays in the kidney for decades.

Source: Institute of Food Safety, Animal Health and Environment “BIOR” / Data-to-Decisions; European Food Safety Authority (EFSA), tolerable weekly intake (TWI)

METALS 21

Methylmercury

CH_3Hg^+ · hydrargyrum





↑ WHAT IT DOES

Affects development of the nervous system

→ WHERE IT IS FOUND

Predatory fish (tuna, pike, perch), Baltic fish, fish oil

HOW OFTEN IT IS FOUND: In 80 % of fish samples

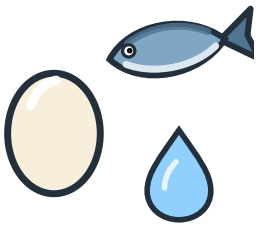
HOW OFTEN IT IS EATEN: Predatory fish, a few times a year

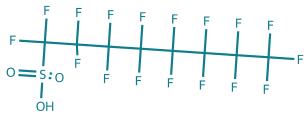
PERSISTENT POLLUTANTS 22

PFAS ("forever chemicals")

$C_8HF_{17}O_3S$

perfluorooctane sulfonic acid (PFOS), a per- and polyfluoroalkyl substance (PFAS)





↑ WHAT IT DOES

Weakens the immune response to vaccination

→ WHERE IT IS FOUND

Fish, eggs, drinking water, offal, food packaging

HOW OFTEN IT IS FOUND: In food, water and almost everyone's blood

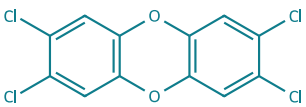
HOW OFTEN IT IS EATEN: Fish, eggs, water, every day

PERSISTENT POLLUTANTS 23

Dioxins and dl-PCBs

$C_{12}H_4Cl_4O_2$ 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD)





↑ WHAT IT DOES

Dioxins and dioxin-like polychlorinated biphenyls (dl-PCBs) affect development and reproduction

→ WHERE IT IS FOUND

Baltic salmon and herring, milk and cheese, meat, eggs, fish oil

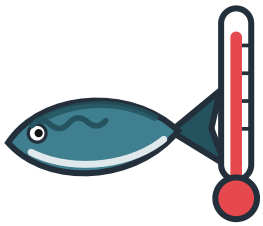
HOW OFTEN IT IS FOUND: Levels in Baltic fish are among the highest in the European Union

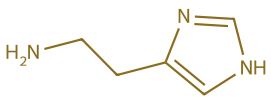
HOW OFTEN IT IS EATEN: Milk, meat, eggs, every day

OTHER 24

Histamine

$C_5H_9N_3$ histamine





↑ WHAT IT DOES

Flushing, headache and palpitations on the same day

→ WHERE IT IS FOUND

Poorly stored fish (herring, sprat, tuna, mackerel), ripened cheese, fish sauce

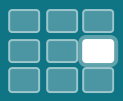
HOW OFTEN IT IS FOUND: In fish it forms only when the fish is kept too warm

HOW OFTEN IT IS EATEN: Fish often; spoiled fish very rarely

RISK ASSESSORS PUT IT HERE

SERIOUS × OFTEN

severity × exposure





PERSISTENT POLLUTANTS

22

PFAS ("forever chemicals")

The carbon-fluorine bond barely breaks down in nature. A large share of Europeans exceed the safe level. The safety value set by the European Food Safety Authority (EFSA) is driven by reduced antibody response to childhood vaccination. Baltic herring and sprat carry a limit 4× higher than fish in general.

HOW MUCH IS TOO MUCH?

The limit is weekly and based on children's vaccine response, the most sensitive effect known.

Source: European Food Safety Authority (EFSA), 2020; Regulation (EU) 2023/915

RISK ASSESSORS PUT IT HERE

SERIOUS × RARELY

severity × exposure





METALS

21

Methylmercury

Detected in 80% of fish samples, the highest detection rate in our dataset. Yet it does not rank high in Latvia, because Latvians eat little fish. "Often detected" is not the same as "high exposure".

HOW MUCH IS TOO MUCH?

The limit is weekly and protects the foetus, not the adult, hence the separate advice in pregnancy.

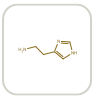
Source: Institute of Food Safety, Animal Health and Environment "BIOR" / Data-to-Decisions; European Food Safety Authority (EFSA)

RISK ASSESSORS PUT IT HERE

MILD × RARELY

severity × exposure





OTHER

24

Histamine

One of the few substances here that cause fast, same-day illness. A cold-chain failure, not a contamination one. Herring and sprat are named as high-risk species in European Union law. Cooking does not destroy it.

HOW MUCH IS TOO MUCH?

The limit is the dose at which no effect appeared in healthy people. Histamine is heat-stable: once formed, cooking will not remove it.

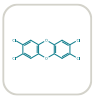
Source: Regulation (EC) No 2073/2005; European Food Safety Authority (EFSA), 2011

RISK ASSESSORS PUT IT HERE

SERIOUS × OFTEN

severity × exposure





PERSISTENT POLLUTANTS

23

Dioxins and dl-PCBs

Average European intake exceeds the safe level. Few cards here can say that. Yet the European Union lets Latvia sell Baltic salmon above the limit domestically, because removing fish from the diet would be worse for health.

HOW MUCH IS TOO MUCH?

The limit is weekly, because dioxins take years to leave. One meal cannot exceed it.

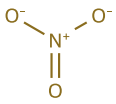
Source: European Food Safety Authority (EFSA), 2026 opinion; Regulation (EU) 2023/915, Article 7

OTHER 25

Nitrate

NO_3^- nitrate ion





WHAT IT DOES

Can form other substances in the body; harmful at high doses

WHERE IT IS FOUND

Leafy vegetables (spinach, rocket, lettuce), beetroot, drinking water, cured meat

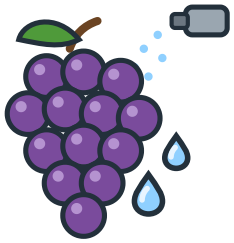
HOW OFTEN IT IS FOUND: About 40 g of rocket already reaches an adult's acceptable daily intake

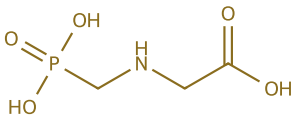
HOW OFTEN IT IS EATEN: Leafy vegetables, almost every day

OTHER 26

Pesticide residues

$C_3H_8NO_5P$ glyphosate, one of more than 400 substances





WHAT IT DOES

Each substance acts in its own way; at high doses some harm the nervous system

WHERE IT IS FOUND

Fruit (grapes, citrus), vegetables, cereals, tea

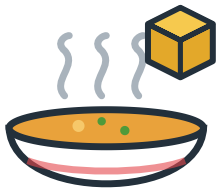
HOW OFTEN IT IS FOUND: In Latvia in 2024, 30 % of samples had quantifiable residues

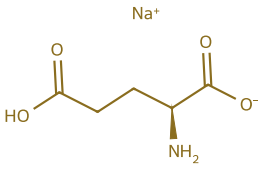
HOW OFTEN IT IS EATEN: Fruit and vegetables, every day

FOOD ADDITIVE 27

Glutamate (E621)

$C_5H_8NNaO_4$ monosodium glutamate (E621)





WHAT IT DOES

In large doses it can cause headaches and raise blood pressure

WHERE IT IS FOUND

Soups and broths, sauces, seasonings, meat products, cakes and pastries

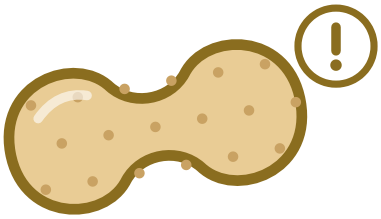
HOW OFTEN IT IS FOUND: In many savoury ready meals; naturally also in cheese and tomatoes

HOW OFTEN IT IS EATEN: Many people, every day

OTHER 28

Peanut allergy

A reaction, not a substance



WHAT IT DOES

In an allergic person, a severe reaction within minutes; can be fatal

WHERE IT IS FOUND

Peanuts, undeclared ingredients, cross-contamination in production

HOW OFTEN IT IS FOUND: Traces can be in products whose ingredients do not list peanuts

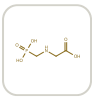
HOW OFTEN IT IS EATEN: Concerns only a few people

RISK ASSESSORS PUT IT HERE

MILD × SOMETIMES

severity × exposure





OTHER

Pesticide residues

26

Europe's #1 public concern. In Latvia in 2024, 70 % of 443 samples had no quantifiable residue at all. A maximum residue level (MRL) is a standard for correct pesticide use, not a health threshold.

HOW MUCH IS TOO MUCH?

The legal maximum residue level is not a health limit: it shows whether the pesticide was used correctly.

Source: European Food Safety Authority (EFSA), pesticide residues report 2024; Eurobarometer, 2025

RISK ASSESSORS PUT IT HERE

MILD × OFTEN

severity × exposure





OTHER

Nitrate

25

Nitrate from food additives is under 5 % of total dietary intake; over 95 % comes from vegetables and water. People worry about the E number on the ham; the intake comes from the salad, and those vegetables are good for you.

HOW MUCH IS TOO MUCH?

This limit is exceeded routinely: about 40 g of rocket reaches an adult's daily figure. The European Food Safety Authority (EFSA) still concludes the benefit of vegetables prevails: what matters is what the nitrate comes with.

Source: European Food Safety Authority (EFSA), 2017 opinion on nitrate

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





OTHER

Peanut allergy

28

Ranked THIRD of 14 food hazards by disease burden in Denmark (2023), above Salmonella, Listeria and cadmium. For most people there is no exposure at all; for an allergic person a mislabelled pack is a same-day risk. Risk is not the same for everyone.

HOW MUCH IS TOO MUCH?

No limit; for an allergic person traces matter, not amounts.

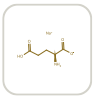
Source: Pires and colleagues, Technical University of Denmark (DTU), Food Control, 2023

RISK ASSESSORS PUT IT HERE

MILD × OFTEN

severity × exposure





FOOD ADDITIVE

Glutamate (E621)

27

In 2022 additives were the most frequently named concern in Latvia, at 44%, higher than the European Union average. Glutamate is one of them: a flavour enhancer in soups, sauces and seasonings. Chemically it is the same as the glutamate in parmesan and tomatoes. In 2017 the European Food Safety Authority (EFSA) set an acceptable daily intake for glutamates. It found that people of any age who eat a lot of foods with these additives, and children with an average diet, may exceed it. An “E” number means a substance has been assessed and approved, not that the amount does not matter.

HOW MUCH IS TOO MUCH?

The acceptable daily intake is 30 mg per kilogram of body weight, about 2 g for a 70 kg person. Larger doses have been linked to headaches and rises in blood pressure and insulin.

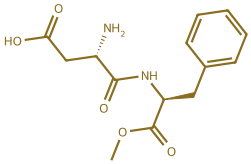
Source: Special Eurobarometer 97.2 (2022), Latvia; European Food Safety Authority (EFSA), 2017 (glutamates, E 620–625); Food Standards Australia New Zealand (FSANZ), 2025

FOOD ADDITIVE 29

Aspartame

$C_{14}H_{18}N_2O_5$ aspartame (E951)





WHAT IT DOES

Assessed; the acceptable daily intake is very high

WHERE IT IS FOUND

Sugar-free drinks, chewing gum, tabletop sweeteners, yoghurt, medicines

HOW OFTEN IT IS FOUND: Only where it is added, not in all processed food

HOW OFTEN IT IS EATEN: Some people, every day

OTHER 30

Microplastics

$(C_2H_4)_n$ polyethylene, one of many polymers





WHAT IT DOES

The health effect is not yet established

WHERE IT IS FOUND

Fish, drinking water, salt, beer, packaging, air

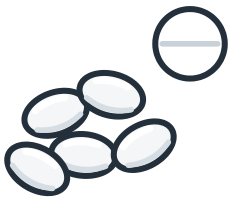
HOW OFTEN IT IS FOUND: In many products, but measuring methods are not yet harmonised

HOW OFTEN IT IS EATEN: Water and salt, every day

FOOD ADDITIVE 31

Titanium dioxide (E171)

TiO_2 titanium dioxide (E171)



WHAT IT DOES

Whether it damages DNA can be neither proven nor ruled out

WHERE IT IS FOUND

Formerly confectionery, chewing gum, icings; still medicines and cosmetics

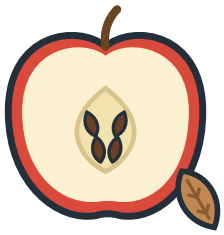
HOW OFTEN IT IS FOUND: Once widely used; banned in food in the European Union since 2022

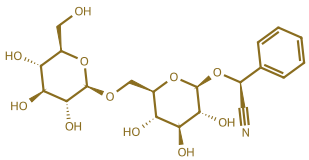
HOW OFTEN IT IS EATEN: No longer in food; in medicines, sometimes

OTHER 32

Cyanogenic glycosides

$C_{20}H_{27}NO_{11}$ amygdalin, releases hydrogen cyanide





WHAT IT DOES

Releases hydrogen cyanide; poisonous at high doses

WHERE IT IS FOUND

Apple seeds, bitter almonds, apricot kernels, linseed, cassava, bamboo shoots

HOW OFTEN IT IS FOUND: So little in apple seeds that hundreds would have to be chewed at once

HOW OFTEN IT IS EATEN: Very rarely, and only a little

RISK ASSESSORS PUT IT HERE

NOT ENOUGH DATA

How well do we know it

?

OTHER

Microplastics

30

A third of Europeans name it a top concern (above bacteria). Yet no authority can quantify either the exposure or the severity: the methods and the toxicology are not there. “We don't know” is not “it's safe”, and not “it's dangerous” either.

HOW MUCH IS TOO MUCH?

No limit, because measurement methods are not yet harmonised.

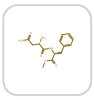
Source: Eurobarometer, 2025; European Food Safety Authority (EFSA), statements

RISK ASSESSORS PUT IT HERE

MILD × SOMETIMES

severity × exposure





FOOD ADDITIVE

Aspartame

29

In July 2023 one body of the World Health Organization (WHO) classified it as “possibly carcinogenic”: a HAZARD statement. Another left the acceptable daily intake (ADI) unchanged: a RISK statement. They answered different questions. Exceeding the ADI means over nine cans a day, every day.

HOW MUCH IS TOO MUCH?

A 70 kg adult would need more than nine cans a day, every day. Even then: aspartame is broken down in the gut into three substances we already eat. A slice of cheese delivers more phenylalanine than a can.

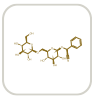
Source: International Agency for Research on Cancer (IARC) and Joint Expert Committee on Food Additives (JECFA), joint statement, July 2023

RISK ASSESSORS PUT IT HERE

MILD × RARELY

severity × exposure





OTHER

Cyanogenic glycosides

32

Apple seeds contain a compound that releases cyanide. A real hazard, yet nobody has ever worried about an apple, because you would have to chew hundreds of seeds at once. A hazard by itself does not mean a high risk. The amount matters too.

HOW MUCH IS TOO MUCH?

This limit is per occasion, not per day: three bitter apricot kernels already exceed it for an adult.

Source: European Food Safety Authority (EFSA), 2019 opinion on cyanogenic glycosides

RISK ASSESSORS PUT IT HERE

NOT ENOUGH DATA

How well do we know it

?

FOOD ADDITIVE

Titanium dioxide (E171)

31

The European Union banned it in food in 2022. Not because harm was proven, but because DNA damage could not be ruled out, so no safe intake could be set. The ban follows from not knowing. Still permitted in medicines and cosmetics.

HOW MUCH IS TOO MUCH?

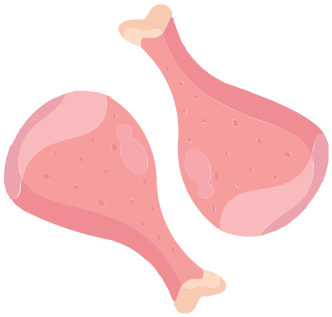
There is no limit: if DNA damage cannot be ruled out, no safe dose can be set, so the European Union banned E171 in food.

Source: European Food Safety Authority (EFSA), 2021; Regulation (EU) 2022/63

WHAT WE DO 33

Undercooked chicken

Campylobacter · Salmonella



WHAT IT DOES

Diarrhoea and fever; many end up in hospital

WHERE IT IS FOUND

Chicken breast and drumsticks, barbecue and kebabs, chicken mince

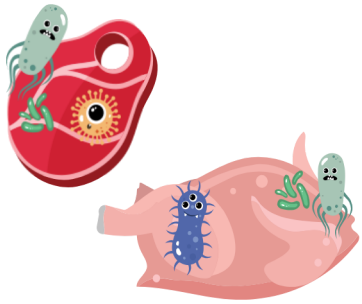
HOW OFTEN IT IS FOUND: In Latvia about a third of fresh chicken samples carry Campylobacter

HOW OFTEN IT IS EATEN: Chicken is eaten almost weekly

WHAT WE DO 34

Home-slaughtered game

Trichinella



WHAT IT DOES

Fever, facial swelling and severe muscle pain for weeks; can damage the heart

WHERE IT IS FOUND

Wild boar, bear meat, home-made game sausage and dried game

HOW OFTEN IT IS FOUND: In Latvia in 1976–2013, 2.5 % of the wild boar tested carried Trichinella

HOW OFTEN IT IS EATEN: A few times a year, and not for everyone

WHAT WE DO 35

One board for meat and salad

transfer from raw meat



WHAT IT DOES

Bacteria end up in food that will never be cooked again; diarrhoea and fever, one in four ends up in hospital

WHERE IT IS FOUND

Cutting board, knife, worktop, barbecue tongs, hands

HOW OFTEN IT IS FOUND: In Latvia about a third of fresh chicken samples carry Campylobacter

HOW OFTEN IT IS EATEN: Almost every time anyone cooks

WHAT WE DO 36

Unwashed hands

norovirus and others



WHAT IT DOES

Vomiting and diarrhoea for a few days

WHERE IT IS FOUND

Before eating, after the toilet, after animals, after raw meat

HOW OFTEN IT IS FOUND: Norovirus outbreaks are regular, and not only because of food

HOW OFTEN IT IS EATEN: Several times a day

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





WHAT WE DO

34

Home-slaughtered game

Freezing does not work here: the Trichinella species found in Latvia are freeze-resistant, and smoking, salting and drying do nothing. Latvia has the highest trichinellosis notification rate in the European Union.

HOW MUCH IS TOO MUCH?

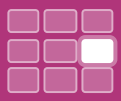
There is no limit: there is a test. One laboratory analysis answers everything.

Source: Kirjušina and colleagues, 2015, Parasites & Vectors; European Centre for Disease Prevention and Control (ECDC), Annual Epidemiological Report 2022

RISK ASSESSORS PUT IT HERE

SERIOUS × OFTEN

severity × exposure





WHAT WE DO

33

Undercooked chicken

Colour is not a thermometer. The only answer is 74 °C in the centre. Latvia reports few cases, not because the bacterium is absent, but because it is rarely looked for: 36.8 % of Latvian fresh chicken samples carried it.

HOW MUCH IS TOO MUCH?

There is no limit: there is a temperature. 74 °C in the centre settles the question.

Source: European Food Safety Authority (EFSA) and European Centre for Disease Prevention and Control (ECDC), One Health Zoonoses Report 2024; Tedersoo and colleagues, 2022, Poultry Science

RISK ASSESSORS PUT IT HERE

MILD × OFTEN

severity × exposure





WHAT WE DO

36

Unwashed hands

Alcohol gel does not work on norovirus: alcohol destroys enveloped viruses, and norovirus has no envelope. Soap and water, not the bottle in your bag.

HOW MUCH IS TOO MUCH?

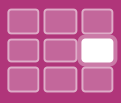
There is no limit: there is soap, water and twenty seconds.

Source: World Health Organization (WHO), Five Keys to Safer Food; European Centre for Disease Prevention and Control (ECDC), 2013, Norovirus in schools and childcare

RISK ASSESSORS PUT IT HERE

SERIOUS × OFTEN

severity × exposure





WHAT WE DO

35

One board for meat and salad

The dangerous object is not the chicken but the board. This card does not change how harmful the bacterium is: it changes where the bacterium ends up.

HOW MUCH IS TOO MUCH?

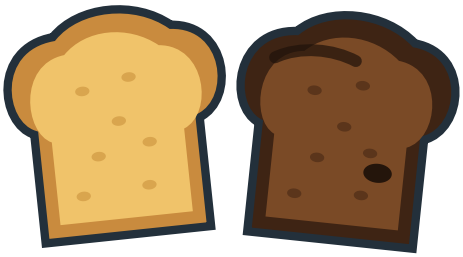
There is no limit: there is an order. Salad first, then meat. Or two boards.

Source: World Health Organization (WHO), Five Keys to Safer Food; European Food Safety Authority (EFSA), 2020, quantitative microbial risk assessment (QMRA) of cross-contamination

WHAT WE DO 37

Cooked too brown

acrylamide



WHAT IT DOES

Damages DNA; considered a probable carcinogen

WHERE IT IS FOUND

Toast, fries, roast potatoes, gingerbread, roasted coffee

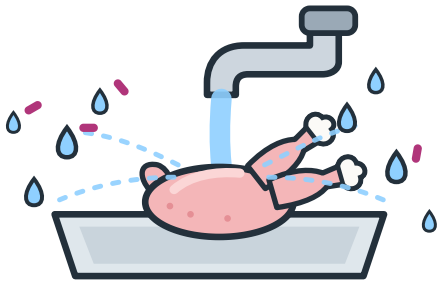
HOW OFTEN IT IS FOUND: Forms whenever food is fried above 120 °C; the darker the crust, the more

HOW OFTEN IT IS EATEN: Almost every day

WHAT WE DO 38

Washing chicken before cooking

splashes carrying Campylobacter



WHAT IT DOES

Bacteria land on surfaces that will never be heated; diarrhoea and fever, one in four ends up in hospital

WHERE IT IS FOUND

Sink, tap, worktop, food standing nearby, clothes

HOW OFTEN IT IS FOUND: In Latvia about a third of fresh chicken samples carry Campylobacter

HOW OFTEN IT IS EATEN: In five European countries, 1–5 in 10 wash it

WHAT WE DO 39

Unwashed salad and berries

norovirus · Salmonella · Shiga toxin-producing Escherichia coli (STEC)



WHAT IT DOES

Vomiting and diarrhoea; occasionally worse

WHERE IT IS FOUND

Leafy salad, strawberries, raspberries, tomatoes, garden greens

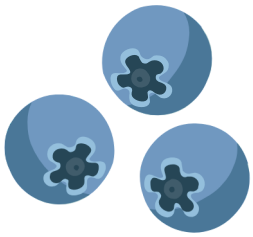
HOW OFTEN IT IS FOUND: Without washing, about ten times more bacteria stay on

HOW OFTEN IT IS EATEN: In season, almost every day

WHAT WE DO 40

Frozen berries straight from the freezer

norovirus · hepatitis A



WHAT IT DOES

Vomiting and diarrhoea; less often hepatitis A, which lasts weeks

WHERE IT IS FOUND

Frozen bilberries and raspberries, smoothies, porridge, desserts

HOW OFTEN IT IS FOUND: Since 2014, over 50 alerts in Europe for norovirus in frozen berries

HOW OFTEN IT IS EATEN: Often in winter, rarely in summer

RISK ASSESSORS PUT IT HERE

SERIOUS × SOMETIMES

severity × exposure



WHAT WE DO

Washing chicken before cooking

38

A hygiene habit that increases risk. Washing does not remove Campylobacter from the bird; it sprays it around the kitchen, up to about half a metre. Only heat makes chicken safe.

HOW MUCH IS TOO MUCH?

Here “wash more” means “spread more”. Only heat makes chicken safe.

Source: Food Standards Agency (FSA) of the United Kingdom, “Don't wash raw chicken”; European Food Safety Authority (EFSA), 2020, quantitative microbial risk assessment (QMRA); SafeConsume survey (Didier et al., 2021)

RISK ASSESSORS PUT IT HERE

SERIOUS × OFTEN

severity × exposure



WHAT WE DO

Cooked too brown

37

“Don’t burn it, lightly brown it” is the official advice. Children have the highest acrylamide exposure of any age group per kilogram of body weight.

HOW MUCH IS TOO MUCH?

Acrylamide has no limit; there are benchmark levels for industry and one rule at home: golden, not dark.

Source: European Food Safety Authority (EFSA), 2015, acrylamide; Regulation (EU) 2017/2158

RISK ASSESSORS PUT IT HERE

MILD × SOMETIMES

severity × exposure



WHAT WE DO

Frozen berries straight from the freezer

40

Freezing does not kill viruses: it preserves them. Note that the official advice says “imported”, yet Finland's 2019 outbreaks came from domestic bilberries. One minute of boiling settles it.

HOW MUCH IS TOO MUCH?

Freezing does not kill the viruses. One minute of boiling does.

Source: Food Safety Authority of Ireland (FSAI), advice on frozen berries; Räsänen and colleagues, 2024

RISK ASSESSORS PUT IT HERE

MILD × OFTEN

severity × exposure



WHAT WE DO

Unwashed salad and berries

39

Washing reduces contamination but does not sterilise, and anything taken up through roots or leaf pores cannot be washed off at all. Re-washing bagged ready-to-eat salad at home is more likely to add contamination than to remove it.

HOW MUCH IS TOO MUCH?

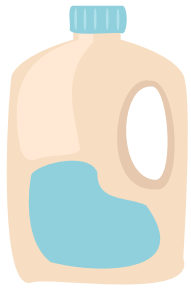
Washing reduces but does not sterilise. Re-washing bagged salad is not worth it.

Source: European Food Safety Authority (EFSA) Panel on Biological Hazards (BIOHAZ), 2014, pathogens in food of non-animal origin

WHAT WE DO 41

Raw milk

Shiga toxin-producing Escherichia coli (STEC) · Campylobacter · Q fever



WHAT IT DOES

Diarrhoea; kidney failure in small children

WHERE IT IS FOUND

Milk vending machines, markets, milk straight from a farm, home cheese

HOW OFTEN IT IS FOUND: In Latvia over five tonnes of raw milk were recalled for Q fever

HOW OFTEN IT IS EATEN: Rarely, but some drink it regularly

WHAT WE DO 42

Mushrooms you picked yourself

amatoxins



WHAT IT DOES

The death cap destroys the liver; fatal in severe cases

WHERE IT IS FOUND

Wild mushrooms, fried mushrooms, pickled mushrooms, mushroom soup

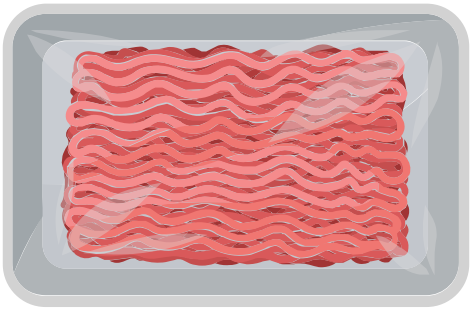
HOW OFTEN IT IS FOUND: Only a few pickers take a poisonous one

HOW OFTEN IT IS EATEN: Many in autumn, few the rest of the year

WHAT WE DO 43

Undercooked burger

Shiga toxin- or verotoxin-producing Escherichia coli (STEC / VTEC)



WHAT IT DOES

Can cause kidney failure, especially in children

WHERE IT IS FOUND

Mince patties, “medium” burgers, home-minced meat, kebabs

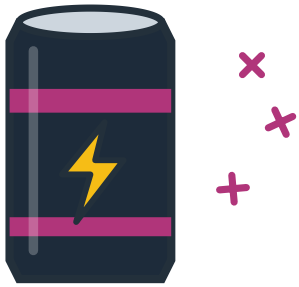
HOW OFTEN IT IS FOUND: On a steak only on the surface; in mince all the way through

HOW OFTEN IT IS EATEN: Several times a month

WHAT WE DO 44

Energy drinks

caffeine



WHAT IT DOES

What it does to teenagers in the long run has not been studied enough

WHERE IT IS FOUND

Energy drinks, colas, coffee drinks, pre-workout mixes

HOW OFTEN IT IS FOUND: One 500 ml can holds 160 mg of caffeine

HOW OFTEN IT IS EATEN: Some teenagers, every day

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





WHAT WE DO

Mushrooms you picked yourself

42

Every folk rule is wrong: the silver spoon does not tarnish, animals do eat poisonous species, boiling does not help, and the death cap tastes fine. The dangerous part is the pause: feeling better on day two is when the liver is being destroyed.

HOW MUCH IS TOO MUCH?

There is no limit: there is identification. If you are not certain, it is not food.

Source: Riga East University Hospital (RAKUS), Toxicology Clinic, 2023; Latvian Poisons and Drug Information Centre

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





WHAT WE DO

Raw milk

41

“Natural” does not mean “safe”. The strangest fact in the deck: you can catch a tick-borne disease by drinking milk: the tick bit the goat, not you. Boiling removes the whole list.

HOW MUCH IS TOO MUCH?

There is no limit: there is boiling. Heated raw milk is a different matter.

Source: European Food Safety Authority (EFSA), 2015, raw drinking milk; Zavadska and colleagues, 2018; Latvian Public Media (LSM), 12.03.2026

RISK ASSESSORS PUT IT HERE

NOT ENOUGH DATA

How well do we know it

?



WHAT WE DO

Energy drinks

44

This card belongs at the question mark. The figure the European Food Safety Authority (EFSA) gives for children is extrapolated from an adult single-dose value because data on children were insufficient. The assessment is being redone. Since 2016 Latvia has banned sales to under-18s.

HOW MUCH IS TOO MUCH?

The figure for children is 3 mg caffeine per kilogram per day, but the European Food Safety Authority (EFSA) itself says the data on children were insufficient.

Source: European Food Safety Authority (EFSA), 2015, caffeine (EFSA Journal 13(5):4102); Saeima, 2016

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × SOMETIMES

severity × exposure





WHAT WE DO

Undercooked burger

43

Same meat, same bacterium, same cook, yet a completely different cell on the matrix. The difference is the mincing machine: a rare steak is safe because the bacteria are only on the surface.

HOW MUCH IS TOO MUCH?

A steak needs only its surface cooked; mince has to be cooked through: 70 °C all the way.

Source: European Centre for Disease Prevention and Control (ECDC), Shiga toxin-producing Escherichia coli (STEC) infection: Annual Epidemiological Report 2023

WHAT WE DO

SET 2 45

Leftovers left out

Bacillus cereus · Clostridium perfringens



WHAT IT DOES

Vomiting or diarrhoea within hours

WHERE IT IS FOUND

Rice, pasta, pilaf, soup or stew left on the stove overnight

HOW OFTEN IT IS FOUND: Beyond 2 hours at room temperature, a toxin can form in the food

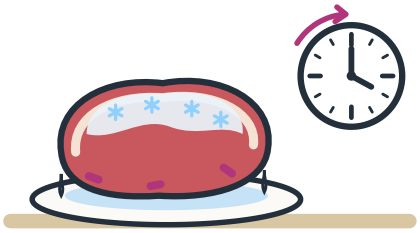
HOW OFTEN IT IS EATEN: In many homes, now and then

WHAT WE DO

SET 2 46

Thawing meat on the counter

bacteria on the surface



WHAT IT DOES

While the middle is still frozen, bacteria multiply on the surface and can cause diarrhoea and vomiting

WHERE IT IS FOUND

Frozen meat, chicken, mince or fish thawing at room temperature

HOW OFTEN IT IS FOUND: Freezing does not kill the bacteria on raw meat; it only puts them to sleep

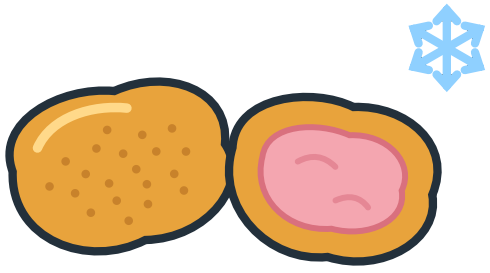
HOW OFTEN IT IS EATEN: Often: about every other person does it

WHAT WE DO

SET 2 47

Frozen breaded chicken

Salmonella



WHAT IT DOES

Salmonellosis: diarrhoea, fever, cramps; one in five went to hospital

WHERE IT IS FOUND

Frozen chicken nuggets, breaded fillets, popcorn chicken, chicken Kiev

HOW OFTEN IT IS FOUND: Salmonella is rare in them: 193 cases in nine countries in three years

HOW OFTEN IT IS EATEN: A children's favourite

WHAT WE DO

SET 2 48

Frozen vegetables, uncooked

Listeria



WHAT IT DOES

Listeriosis: dangerous in pregnancy, old age and for the frail

WHERE IT IS FOUND

Frozen corn, peas, spinach and vegetable mixes in salads or smoothies

HOW OFTEN IT IS FOUND: In a 2015–2018 outbreak, 47 people fell ill in five countries

HOW OFTEN IT IS EATEN: Usually cooked, sometimes not

RISK ASSESSORS PUT IT HERE

MILD × OFTEN

severity × exposure





WHAT WE DO

46

Thawing meat on the counter

Freezing does not kill bacteria; it pauses them. While the middle is still hard, the surface spends hours in the warm, where bacteria grow fastest. The European Food Safety Authority (EFSA) lists it among thawing habits of concern. Safe ways: in the fridge, in cold water in a sealed bag, or in the microwave.

HOW MUCH IS TOO MUCH?

The limit is temperature: in a fridge below 5 °C bacteria barely grow.

Source: Food Standards Agency (FSA) of the United Kingdom, Food and You 2 (2024); Bensa and colleagues, 2026 (Slovenia); European Food Safety Authority (EFSA), 2021 (date marking, Part 2); Centre for Disease Prevention and Control of Latvia (SPKC), 2022

RISK ASSESSORS PUT IT HERE

MILD × SOMETIMES

severity × exposure





WHAT WE DO

45

Leftovers left out

Reheating does not fix it. Bacillus cereus spores survive cooking, wake up in warm food and make a vomiting toxin that heat does not destroy. The advice is about time: no more than two hours at room temperature, then the fridge; freeze what you will keep longer than 2–3 days.

HOW MUCH IS TOO MUCH?

The limit is time: no more than 2 hours at room temperature, then into the fridge.

Source: French Agency for Food, Environmental and Occupational Health & Safety (ANSES); European Food Safety Authority (EFSA), 2016 (Bacillus cereus); Centre for Disease Prevention and Control of Latvia (SPKC), 2022

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





WHAT WE DO

48

Frozen vegetables, uncooked

From 2015 to 2018, 47 people in five countries caught listeriosis from frozen corn made in Hungary; nine died, and 23 of the cases were in Finland. Listeria survives freezing, so cook frozen vegetables even for salads and smoothies.

HOW MUCH IS TOO MUCH?

The limit is on the pack: “cook before eating”.

Source: European Centre for Disease Prevention and Control (ECDC) and European Food Safety Authority (EFSA), rapid outbreak assessment, 2018; EFSA, 2020

RISK ASSESSORS PUT IT HERE

SERIOUS × RARELY

severity × exposure





WHAT WE DO

47

Frozen breaded chicken

Between 2018 and 2020, 193 people in nine European countries caught salmonellosis from frozen breaded chicken made in Poland, and one in five went to hospital. The crumb is fried; the meat inside is raw. The cooking time on the pack is an instruction, not a suggestion.

HOW MUCH IS TOO MUCH?

There is no limit, only the instructions on the pack: how long and how hot.

Source: European Centre for Disease Prevention and Control (ECDC) and European Food Safety Authority (EFSA), rapid outbreak assessment, 2021; Food Standards Agency (FSA) of the United Kingdom, 2022; Epidemiology and Infection (Canada)

WHAT WE DO

SET 2 49

Raw dough

Shiga toxin-producing Escherichia coli (STEC) · Salmonella



WHAT IT DOES

E. coli (STEC): bloody diarrhoea; in children it can damage the kidneys

WHERE IT IS FOUND

Raw cookie and cake dough, pancake batter, a licked spoon, flour play dough

HOW OFTEN IT IS FOUND: German tests found live STEC in one in ten ready-made doughs

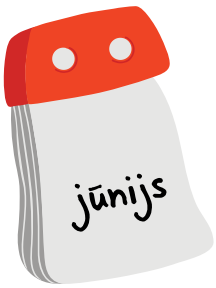
HOW OFTEN IT IS EATEN: In Germany one in five, at least once a month

WHAT WE DO

SET 2 50

Eating food past its “use by” date

Listeria



WHAT IT DOES

Listeria keeps growing in the fridge; listeriosis can be fatal

WHERE IT IS FOUND

Mince, cold-smoked fish, ready salads, cold cuts and soft cheese past the date

HOW OFTEN IT IS FOUND: Listeria is in a small share of products; past the date there may be more

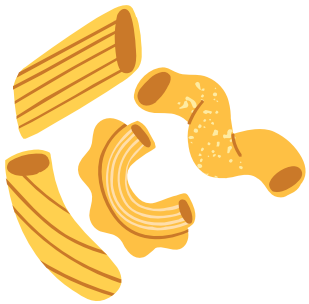
HOW OFTEN IT IS EATEN: Rarely, so as not to waste it

WHAT WE DO

SET 2 51

Eating food past its “best before” date

a quality date



WHAT IT DOES

Usually nothing: taste or texture may change

WHERE IT IS FOUND

Pasta, rice, grains, tinned food, chocolate, crisps, frozen vegetables

HOW OFTEN IT IS FOUND: Only when the pack is damaged or the food was stored wrongly

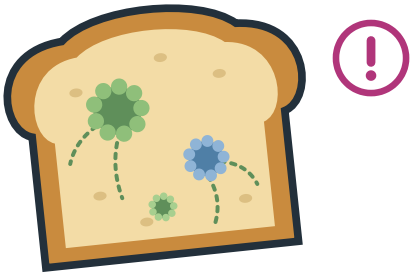
HOW OFTEN IT IS EATEN: In almost every home

WHAT WE DO

SET 2 52

Scraping off the mould

mycotoxins



WHAT IT DOES

Nobody has measured how much toxin you take in when you scrape off the mould

WHERE IT IS FOUND

Bread, jam, yogurt, soft fruit and berries with the mould scraped or cut off

HOW OFTEN IT IS FOUND: Toxins spread beyond the visible spot

HOW OFTEN IT IS EATEN: In Germany one in four does it with jam

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





WHAT WE DO

Eating food past its “use by” date

50

Two labels that look alike but belong in very different squares. “Use by” is a safety date: European Union law says food past it is deemed unsafe. Listeria grows even at 2 °C, and the European Food Safety Authority (EFSA) estimates that a third of listeriosis cases come from food prepared and stored at home. Freeze it before the date, not after.

HOW MUCH IS TOO MUCH?

The date itself is the limit. After it, the food may not be sold and should not be eaten.

Source: Regulation (EU) No 1169/2011, Article 24; European Food Safety Authority (EFSA), 2018, 2020; Eurobarometer 425 (2015)

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × SOMETIMES

severity × exposure





WHAT WE DO

Raw dough

49

Flour looks finished but is a raw crop: bacteria cannot grow in dry flour, yet they survive, and only baking kills them. It is thought of as an American problem, but in 2021 German testing found live Shiga toxin-producing Escherichia coli (STEC) in 35 of 335 ready-made doughs and baking mixes.

HOW MUCH IS TOO MUCH?

There is no limit: only baking makes dough safe.

Source: German Federal Institute for Risk Assessment (BfR), 2023; German Federal Office of Consumer Protection and Food Safety (BVL), 2022; European Centre for Disease Prevention and Control (ECDC), 2023

RISK ASSESSORS PUT IT HERE

NOT ENOUGH DATA

How well do we know it

?



WHAT WE DO

Scraping off the mould

52

Mould is not just the spot on the surface: its threads grow into soft bread and jam, and toxins spread further. Heat kills the mould but not its toxins. And moulds differ: the blue in Roquefort and the white rind on Camembert are grown on purpose and safe; wild mould on bread is not.

HOW MUCH IS TOO MUCH?

There is no limit: cutting it off does not protect against mycotoxins.

Source: Centre for Disease Prevention and Control of Latvia (SPKC), 2022; German Federal Institute for Risk Assessment (BfR), 2024, 2026; Mattilsynet

RISK ASSESSORS PUT IT HERE

MILD × OFTEN

severity × exposure





WHAT WE DO

Eating food past its “best before” date

51

“Best before” is about quality, not safety. In Latvia shops may not sell such food, but they may donate it: pasta and chocolate up to two months past the date, bread two days.

HOW MUCH IS TOO MUCH?

There is no limit, just common sense: an intact pack, the look, the smell.

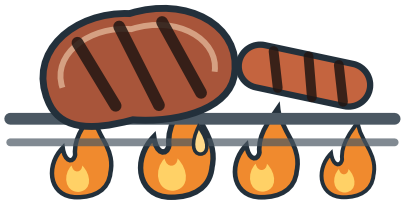
Source: Regulation (EU) No 1169/2011; Cabinet Regulation No 514; Ministry of Climate and Energy of Latvia (KEM), 2025

WHAT WE DO

SET 2 53

Charred barbecue meat

polycyclic aromatic hydrocarbons (PAHs) · heterocyclic amines



WHAT IT DOES

PAHs and heterocyclic amines damage DNA; the harm builds up over years

WHERE IT IS FOUND

Shashlik, sausages, chops with black edges, meat over flaring fat

HOW OFTEN IT IS FOUND: PAHs come from smoke; heterocyclic amines form in the meat above 150 °C

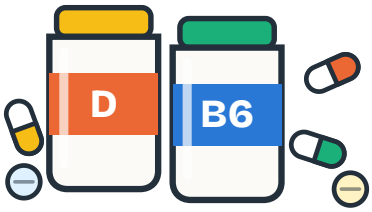
HOW OFTEN IT IS EATEN: In summer, several times a month

WHAT WE DO

SET 2 54

Several supplements at once

vitamin D · vitamin B6



WHAT IT DOES

Too much vitamin D harms the kidneys; too much vitamin B6 harms the nerves

WHERE IT IS FOUND

Vitamin D, multivitamins, B vitamins, magnesium, green tea extract

HOW OFTEN IT IS FOUND: A too-high dose comes when one nutrient is in several products

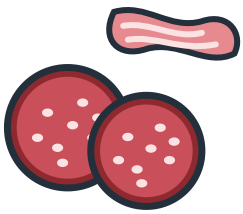
HOW OFTEN IT IS EATEN: Many people daily, but too-high doses are rare

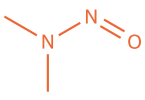
PROCESS CONTAMINANTS

SET 2 55

Nitrosamines

$C_2H_6N_2O$ N-nitrosodimethylamine (NDMA)





WHAT IT DOES

Damage DNA; cause liver tumours in animals

WHERE IT IS FOUND

Sausages, ham, bacon, smoked fish, beer, cocoa, pickled vegetables

HOW OFTEN IT IS FOUND: Forms whenever nitrite reacts with proteins, especially when heated

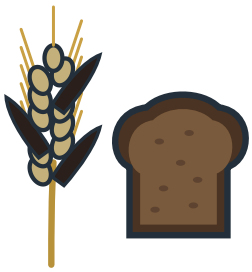
HOW OFTEN IT IS EATEN: Almost every day

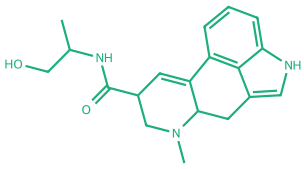
MYCOTOXINS

SET 2 56

Ergot alkaloids

$C_{19}H_{23}N_3O_2$ ergometrine, one of the ergot alkaloids





WHAT IT DOES

Narrows blood vessels; the severe poisonings of the Middle Ages no longer happen in Europe

WHERE IT IS FOUND

Rye bread, dark rye bread, rye flour, rye flakes

HOW OFTEN IT IS FOUND: Ergot still turns up in Latvian rye fields today

HOW OFTEN IT IS EATEN: Many people eat dark rye bread every day

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





WHAT WE DO

54

Several supplements at once

A supplement is food, not medicine, and in the European Union there is no safety check before it goes on sale. Trouble starts when the same nutrient comes from several products, or big doses are taken “just in case”. The Netherlands has 238 reports of nerve damage linked to vitamin B6; the European Food Safety Authority (EFSA) set its upper level at 12 mg a day in 2023.

HOW MUCH IS TOO MUCH?

The upper level is an amount not to exceed long-term; children's are lower: vitamin D 50 µg a day at ages 1–10.

Source: European Food Safety Authority (EFSA), 2023 (vitamins D and B6); Lareb, 2024; Institute of Food Safety, Animal Health and Environment “BIOR”, 2025

RISK ASSESSORS PUT IT HERE

SERIOUS × SOMETIMES

severity × exposure





WHAT WE DO

53

Charred barbecue meat

Barbecuing makes two groups of chemicals: polycyclic aromatic hydrocarbons (PAHs), when fat drips into the flames and smoke coats the meat, and heterocyclic amines, which form inside meat above 150 °C, even in a pan. The fixes are the same: nothing dripping onto the coals, brown not black, cut off the charred bits.

HOW MUCH IS TOO MUCH?

Grilled meat that is sold may hold at most 5 µg of benzo(a)pyrene per kg; heterocyclic amines have no limit.

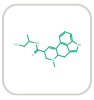
Source: European Food Safety Authority (EFSA), 2008; International Agency for Research on Cancer (IARC); German Federal Institute for Risk Assessment (BfR), 2026; Regulation (EU) 2023/915

RISK ASSESSORS PUT IT HERE

SERIOUS × SOMETIMES

severity × exposure





MYCOTOXINS

56

Ergot alkaloids

In medieval Western Europe, ergot poisoning was called “St Anthony's fire”. “Fire” because hands and feet burned with pain. Some people also hallucinated and had seizures. “St Anthony's” because monks of the Order of St Anthony cared for the sick. They also fed them bread from clean grain.

Rye bread is still a daily food in Latvia, and ergot still turns up in Latvian rye fields. In July 2024 the European Union halved the limit for rye flour to 250 µg/kg.

HOW MUCH IS TOO MUCH?

Rye flour: at most 250 µg/kg (since July 2024); baby food: 20 µg/kg.

Source: European Food Safety Authority (EFSA), 2012, 2017; Regulation (EU) 2021/1399; Latvia University of Agriculture (LLU), 2008

RISK ASSESSORS PUT IT HERE

SERIOUS × OFTEN

severity × exposure





PROCESS CONTAMINANTS

55

Nitrosamines

Nitrosamines form when nitrite, the preservative that keeps cured meat pink and prevents botulism, reacts with proteins, especially when heated. The European Food Safety Authority (EFSA) concluded in 2023 that exposure raises a health concern for all age groups; since 9 October 2025 the European Union allows at most 80 mg/kg of added nitrite in most meat products.

HOW MUCH IS TOO MUCH?

Added nitrite in meat products: at most 80 mg/kg (since October 2025).

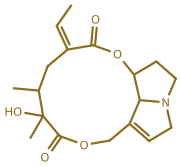
Source: European Food Safety Authority (EFSA), 2023 (N-nitrosamines in food); Regulation (EU) 2023/2108

OTHER SET 2 57

Pyrrolizidine alkaloids

$C_{18}H_{25}NO_5$ senecionine, one of the pyrrolizidine alkaloids





WHAT IT DOES

Severely damages the liver; damages DNA

WHERE IT IS FOUND

Herbal teas, tea, honey, herbs, herbal supplements

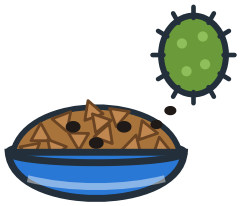
HOW OFTEN IT IS FOUND: The BIOR institute found them in 7 of 15 Latvian herbal teas (2020)

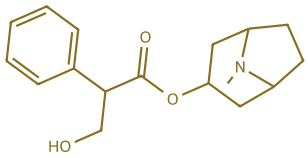
HOW OFTEN IT IS EATEN: Herbal-tea lovers, every day

OTHER SET 2 58

Tropane alkaloids

$C_{17}H_{23}NO_3$ atropine





WHAT IT DOES

Dry mouth, racing heart, confusion, hallucinations

WHERE IT IS FOUND

Buckwheat, buckwheat flour, millet, sorghum, maize, herbal teas

HOW OFTEN IT IS FOUND: Only when thorn-apple grows in the field and its seeds get into the harvest

HOW OFTEN IT IS EATEN: Buckwheat often, contaminated batches rarely

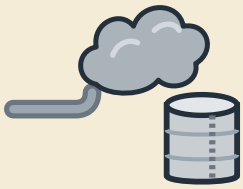
METALS SET 2 59

A CASE FROM HISTORY Europe, 1970s

One real case from the past, assessed for the people who lived then and there.

Lead from petrol

plumbum



82
Pb
207,2

WHAT IT DOES

Permanently harms children's brain development: lower intelligence quotient (IQ), changed behaviour

WHERE IT IS FOUND

All everyday food, the air, cans with lead solder

HOW OFTEN IT IS FOUND: Everywhere, as road traffic in Europe emitted 119,000 t of lead in 1975

HOW OFTEN IT IS EATEN: Everyone, every day, including by breathing

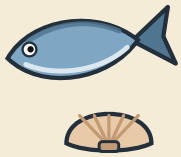
METALS SET 2 60

A CASE FROM HISTORY Japan, 1956

One real case from the past, assessed for the people who lived then and there.

Methylmercury in Minamata

CH_3Hg^+ · hydrargyrum



80
Hg
200,6

WHAT IT DOES

Severely damages the nervous system; can kill; harms children before birth

WHERE IT IS FOUND

Fish and shellfish from Minamata Bay

HOW OFTEN IT IS FOUND: So much mercury that from about 1953 cats that ate the fish went mad and died

HOW OFTEN IT IS EATEN: Fishing families ate fish every day, often three times a day

RISK ASSESSORS PUT IT HERE

SERIOUS × RARELY

severity × exposure



OTHER

58

Tropane alkaloids

Thorn-apple is a weed whose seeds contain atropine and scopolamine, the drugs doctors use to dilate pupils. When it grows in a buckwheat field, its seeds end up in the harvest. In Slovenia in 2003, 73 people fell ill after eating buckwheat products; since 2022 the European Union has a limit for buckwheat.

HOW MUCH IS TOO MUCH?

Buckwheat: at most 10 µg/kg in the European Union (atropine and scopolamine together).

Source: European Food Safety Authority (EFSA), 2013, 2018; Regulation (EU) 2021/1408; Austrian Agency for Health and Food Safety (AGES)

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × SOMETIMES

severity × exposure



OTHER

57

Pyrrolizidine alkaloids

Plants make pyrrolizidine alkaloids to fend off insects; they get into herbal tea when weeds such as ragwort are harvested with the crop. In 2020 the Institute of Food Safety, Animal Health and Environment “BIOR” found them in almost half of the Latvian herbal teas tested, one yarrow tea close to the European Union limit. One cup means nothing; the concern is for people who drink a lot of the same tea every day for years.

HOW MUCH IS TOO MUCH?

Herbal teas: at most 200 µg/kg in the European Union; honey has no limit.

Source: European Food Safety Authority (EFSA), 2017; Institute of Food Safety, Animal Health and Environment “BIOR”, 2020; Regulation (EU) 2020/2040

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × OFTEN

for a fishing family on Minamata Bay in the 1950s



80 Hg

METALS · HISTORY · JAPAN, 1956

60

Methylmercury in Minamata

In 1956 a new disease was officially reported in Minamata, among people who ate fish from the bay. Since 1932 a chemical plant had discharged waste with methylmercury into the sea, and it built up in fish and shellfish, which fishing families ate daily. Japan has officially recognised 3,000 patients, 716 of them from a second outbreak in Niigata. Discharge stopped in 1968, and the Minamata Convention on Mercury entered into force in 2017.

HOW MUCH IS TOO MUCH?

In 1957 fish sales were not banned. Today Regulation (EU) 2023/915 caps mercury in fish: 0.50 mg/kg in general, 1.0 mg/kg in large predatory fish.

Source: Ministry of the Environment of Japan (2002, 2025); European Environment Agency (EEA), “Late lessons from early warnings II”, Chapter 5 (2013); Takaoka and colleagues, 2018

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × OFTEN

for a child in Europe in the 1970s



82 Pb

METALS · HISTORY · EUROPE, 1970S

59

Lead from petrol

In the 1970s lead was everywhere: cars blew it out (road traffic in Europe emitted 119,000 t in 1975), and it was in the solder of food cans. In the United Kingdom the diet held 110 µg of lead a day in 1976 and 6 µg in 2006. The European Union cut lead in petrol step by step and banned leaded petrol from 1 January 2000. Lead has not gone, so the card “Lead” still sits at “often”, but the amounts are many times smaller.

HOW MUCH IS TOO MUCH?

In 1978 the European Economic Community (EEC) allowed petrol to hold up to 0.4 g of lead per litre. Today Regulation (EU) 2023/915 caps lead in food, for example 0.20 mg/kg in cereals.

Source: European Food Safety Authority (EFSA), 2010 (lead in food); European Environment Agency (EEA), “Late lessons from early warnings II”, Chapter 3 (2013); Stroh and colleagues, 2009; Directive 98/70/EC

METALS SET 2 61

A CASE FROM HISTORY Japan, 1910s–1970s
One real case from the past, assessed for the people who lived then and there.

Cadmium in Toyama rice

cadmium





WHAT IT DOES

Damages the kidneys, then softens the bones until they break again and again

WHERE IT IS FOUND

Rice from paddies irrigated with Jinzū river water, and river water to drink

HOW OFTEN IT IS FOUND: In every harvest, as the soil of 1,686.2 ha of paddies was polluted

HOW OFTEN IT IS EATEN: Rice from their own paddies, every day, for decades

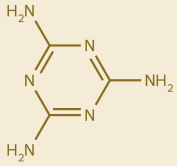
OTHER SET 2 62

A CASE FROM HISTORY China, 2008
One real case from the past, assessed for the people who lived then and there.

Melamine in infant formula

$C_3H_6N_6$ melamine





WHAT IT DOES

Forms stones in the kidneys and urinary tract; in babies the kidney damage can be fatal

WHERE IT IS FOUND

Infant formula and milk products

HOW OFTEN IT IS FOUND: 69 of 491 batches; the most contaminated brand, Sanlu, averaged 1,212 mg/kg

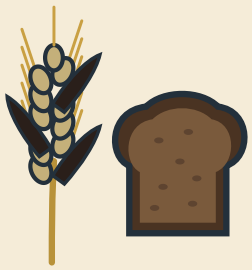
HOW OFTEN IT IS EATEN: For the babies it was the main food, for some the only one, several times a day

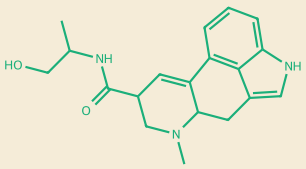
MYCOTOXINS SET 2 63

A CASE FROM HISTORY Urals, 1926–1927
One real case from the past, assessed for the people who lived then and there.

Ergotism epidemic

$C_{19}H_{23}N_3O_2$ ergometrine, one of the ergot alkaloids





WHAT IT DOES

Painful cramps, fits and psychoses; can kill

WHERE IT IS FOUND

Rye bread, and kvass and beer from the same rye

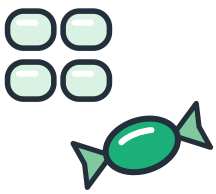
HOW OFTEN IT IS FOUND: 45% of 457 flour samples held a lot of ergot

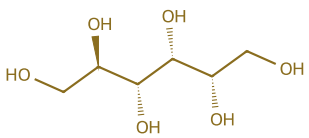
HOW OFTEN IT IS EATEN: Everyone, several times a day, all autumn and winter

FOOD ADDITIVE SET 2 64

Polyols (sugar alcohols)

$C_6H_{14}O_6$ sorbitol (E420), one of the polyols





WHAT IT DOES

In large amounts they cause diarrhoea

WHERE IT IS FOUND

Sugar-free chewing gum, sugar-free sweets and chocolate, sweeteners

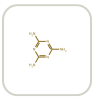
HOW OFTEN IT IS FOUND: Mostly in sugar-free products

HOW OFTEN IT IS EATEN: Some people, every day

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × OFTEN

for a baby fed on a contaminated brand of formula in China in 2008



OTHER · HISTORY · CHINA, 2008

62

Melamine in infant formula

In China in 2008 melamine was added to watered-down milk. It is rich in nitrogen, and the usual test read it as protein, so the milk looked complete. Products of 22 companies carried it. For formula-fed babies it was the main food, for some the only one: 294,000 were affected and six died. The European Union banned such imports from China that year and now sets limits for melamine in food.

HOW MUCH IS TOO MUCH?

After the crisis the European Union banned such imports from China in 2008. Today the limit is 2.5 mg/kg in food, and in infant formula 1 mg/kg as powder and 0.15 mg/kg as liquid.

Source: World Health Organization (WHO), 2009 (melamine and cyanuric acid); Commission Decision 2008/798/EC; Regulation (EU) 2023/915

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × OFTEN

for a long-term resident of the Jinzū river basin in the mid-20th century

48 Cd

METALS · HISTORY · JAPAN, 1910S–1970S

61

Cadmium in Toyama rice

From about 1912 people in the Jinzū river basin in Japan suffered pain all over the body. The disease was called “itai-itai”, “it hurts, it hurts”. The cause was cadmium from the Kamioka mine: irrigation water carried it into the rice paddies and the rice, and people also drank the river water. In all, 201 patients were officially recognised. In 1968 the government named the cause; the paddies were rebuilt with clean soil in 1979–2012.

HOW MUCH IS TOO MUCH?

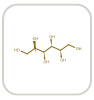
Japan set a limit for rice (1 mg/kg) only in 1970. Today Regulation (EU) 2023/915 caps cadmium in rice at 0.15 mg/kg.

Source: Toyama Prefecture; Ministry of the Environment of Japan (2025); Kaji, 2012

RISK ASSESSORS PUT IT HERE

MILD × SOMETIMES

severity × exposure



FOOD ADDITIVE

64

Polyols (sugar alcohols)

Polyols, or sugar alcohols, are sweeteners: sorbitol, xylitol, erythritol and others. They taste sweet like sugar and go into chewing gum, sweets and other sugar-free products. The gut absorbs them only in part, so in large amounts they cause diarrhoea. A food with more than 10 % added polyols must warn that excessive consumption may have a laxative effect. In 2023 the European Food Safety Authority (EFSA) set a daily limit for erythritol based on this effect and found that people with a high intake exceed it.

HOW MUCH IS TOO MUCH?

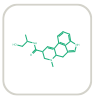
For erythritol the daily limit is 0.5 g per kilogram of body weight: 10 g for a 20 kg child, 35 g for a 70 kg adult.

Source: Regulation (EU) No 1169/2011, Annex III; European Food Safety Authority (EFSA), 2023 (erythritol, E 968)

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × OFTEN

for a peasant in the Urals from autumn 1926 to spring 1927



MYCOTOXINS · HISTORY · URALS, 1926–1927

63

Ergotism epidemic

The summer of 1926 in the Urals was cold and wet, and rye flowered for a long time. Ergot spread: in some fields every ear was affected, and 45% of flour samples held a lot of it. People baked bread and brewed kvass from that rye. Four districts registered more than 10,000 cases. Today grain is cleaned and sorted and the European Union sets limits, but ergot still grows in Latvian rye fields.

HOW MUCH IS TOO MUCH?

A ban on milling rye with a lot of ergot was ignored even at state mills. Today the European Union limit for unprocessed rye is 0.2 g of ergot per kilogram.

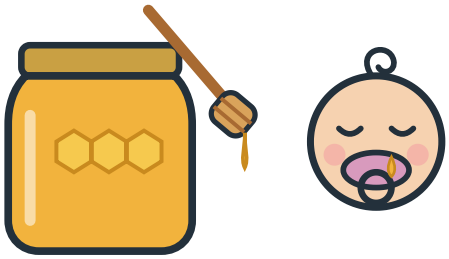
Source: Maksudov, Kazan Medical Journal, 1927 (in Russian); Regulation (EU) 2021/1399; Bankina and colleagues, 2008, Latvia University of Agriculture (LLU)

WHAT WE DO

SET 2 65

Honey for babies

Clostridium botulinum (spores)



WHAT IT DOES

The baby's muscles weaken: constipation, weak sucking, trouble breathing

WHERE IT IS FOUND

Honey for a baby under 12 months: on a spoon, in porridge, on a dummy or a finger

HOW OFTEN IT IS FOUND: Spores in honey are uncommon, but in a baby's gut they can germinate

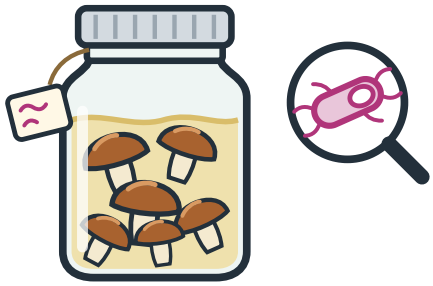
HOW OFTEN IT IS EATEN: To soothe a baby or sweeten its food

WHAT WE DO

SET 2 66

Home-canned food

Clostridium botulinum (toxin)



WHAT IT DOES

Blurred vision, trouble speaking and swallowing, paralysis

WHERE IT IS FOUND

Home-canned mushrooms, beans or beetroot; garlic or herbs in oil

HOW OFTEN IT IS FOUND: Only in jars canned the wrong way, with no acid

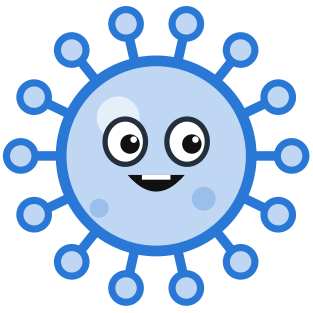
HOW OFTEN IT IS EATEN: Many eat home-canned food, but botulism is rare

MICRO-ORGANISMS

SET 2 67

Hepatitis E virus

Hepatitis E virus (genotype 3)



WHAT IT DOES

Often no signs; sometimes tiredness, nausea, jaundice

WHERE IT IS FOUND

Raw or undercooked pork, pig liver, liver sausage and pâté, wild boar

HOW OFTEN IT IS FOUND: Relatively common in pork; thorough cooking destroys it

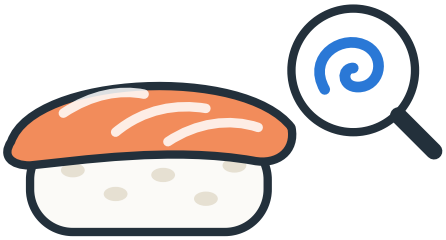
HOW OFTEN IT IS EATEN: Pork often, undercooked pork less often

MICRO-ORGANISMS

SET 2 68

Anisakis: fish parasites

Anisakis (roundworm larvae)



WHAT IT DOES

Sharp stomach pain like an ulcer; allergy in some

WHERE IT IS FOUND

Raw or lightly salted fish, sushi, ceviche, home-salted herring from unfrozen fish

HOW OFTEN IT IS FOUND: Larvae are in many sea fish, but only live larvae can do harm

HOW OFTEN IT IS EATEN: Live larvae rarely: restaurants freeze fish meant to be eaten raw

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





WHAT WE DO

66

Home-canned food

Clostridium botulinum spores live in soil and easily reach mushrooms and vegetables. Ordinary boiling does not kill them. In a sealed jar without air or acid the spores can germinate and make toxin. Mushrooms, beans and beetroot are not acidic.

In 2022 Europe had 84 confirmed botulism cases, most often from canned food, often home-made. Latvia reported no confirmed case in 2015–2022. The German Federal Institute for Risk Assessment (BfR) advises against storing home-made vegetables or herbs in oil.

HOW MUCH IS TOO MUCH?

Throw away a bulging or doubtful jar. Do not keep garlic or herbs in oil at home. Boiling destroys the toxin: 5 minutes above 85 °C.

Source: European Centre for Disease Prevention and Control (ECDC), annual epidemiological report on botulism, 2022; World Health Organization (WHO), botulism, 2023; German Federal Institute for Risk Assessment (BfR), 2016

RISK ASSESSORS PUT IT HERE

VERY SERIOUS × RARELY

severity × exposure





WHAT WE DO

65

Honey for babies

Honey can carry spores of Clostridium botulinum. In an adult gut they do not develop. A baby's gut flora is not yet mature, so the spores germinate there and release a toxin that paralyses muscles. The first sign is usually constipation, then weak sucking and poor head control.

The illness is very rare but severe: it usually means a long hospital stay, often with breathing support. Deaths in Western countries are extremely rare. The World Health Organization (WHO) and the French food agency (ANSES) advise no honey at all before 12 months, not even on a dummy or a finger.

HOW MUCH IS TOO MUCH?

No honey at all before 12 months: not on a spoon, in porridge or on a dummy. Boiling does not kill the spores.

Source: World Health Organization (WHO), botulism fact sheet (2023); French Agency for Food, Environmental and Occupational Health and Safety (ANSES), 2013; Harris and Dabritz, 2024; European Centre for Disease Prevention and Control (ECDC), botulism, 2022

RISK ASSESSORS PUT IT HERE

SERIOUS × RARELY

severity × exposure





MICRO-ORGANISMS

68

Anisakis: fish parasites

Anisakis are roundworm larvae living in sea fish. The European Food Safety Authority (EFSA) found that no sea fishing area is free of them. A Polish study in 2016 found them in 14 % of Baltic herring. Brief salting, marinating and cold smoking do not reliably kill them.

Freezing (-20 °C for 24 hours) or heating to 60 °C for at least 1 minute does. The allergens remain. Spain, with the highest rate in Europe, has an estimated 7,700–8,320 cases a year, mostly from marinated anchovies.

HOW MUCH IS TOO MUCH?

Freeze fish to be eaten raw or lightly salted first: -20 °C for 24 hours or -15 °C for 4 days. Or cook to 60 °C in the centre.

Source: European Food Safety Authority (EFSA), parasites in fishery products, 2010 and 2024; French Agency for Food, Environmental and Occupational Health and Safety (ANSES), Anisakis, 2008; Bao et al., 2017; Różycki et al. (Baltic Sea, 2016 samples)

RISK ASSESSORS PUT IT HERE

MILD × SOMETIMES

severity × exposure





MICRO-ORGANISMS

67

Hepatitis E virus

In Europe most human infections are hepatitis E virus genotype 3, and the main source is raw or undercooked pork and liver, and wild boar. Freezing does not kill the virus; only thorough cooking does.

Most infected people notice nothing. In Germany and France about one person in five has antibodies, so they have had the infection. People with liver disease or weak immunity can fall seriously ill. Europe reported more than 21,000 cases and 28 deaths in 2005–2015.

HOW MUCH IS TOO MUCH?

Cook pork, liver and wild boar through. Freezing does not kill the virus. With liver disease, avoid raw pork sausages.

Source: European Food Safety Authority (EFSA), hepatitis E virus, 2017; European Centre for Disease Prevention and Control (ECDC), hepatitis E in Europe 2005–2015, 2017; German Federal Institute for Risk Assessment (BfR), questions and answers on hepatitis E virus, 2016; Szabo et al., 2015

OTHER SET 2 69

Chemicals from packaging and tableware

Melamine · formaldehyde · bisphenols



WHAT IT DOES

Substances from plastics and coatings pass into food; the limits are set for a whole lifetime

WHERE IT IS FOUND

Hot food in plastic containers, cans, bamboo and melamine cups, nylon spoons

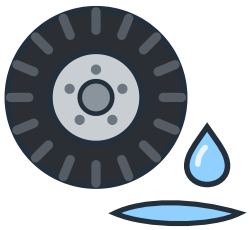
HOW OFTEN IT IS FOUND: Very little passes into food, and goods that exceed the limits are withdrawn

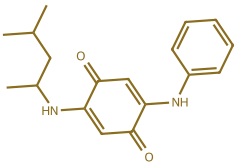
HOW OFTEN IT IS EATEN: Every day: nearly all food has been packed

OTHER SET 2 70

6PPD-quinone

$C_{18}H_{22}N_2O_2$ 6PPD-quinone





WHAT IT DOES

A tyre-wear chemical that kills salmon; the effect on people is unknown

WHERE IT IS FOUND

Leafy vegetables and fish have been studied; it is also found in road runoff and house dust

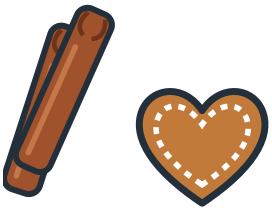
HOW OFTEN IT IS FOUND: Found in human urine and blood, but the source and dose are unknown

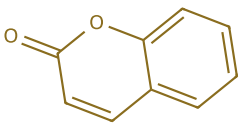
HOW OFTEN IT IS EATEN: How much comes with food is unknown

OTHER SET 2 71

Cinnamon and coumarin

$C_9H_6O_2$ Coumarin





WHAT IT DOES

In large doses it briefly affects the liver of sensitive people; it passes when they stop

WHERE IT IS FOUND

Gingerbread, cinnamon buns, porridge or rice with cinnamon, apple cake, ground cinnamon

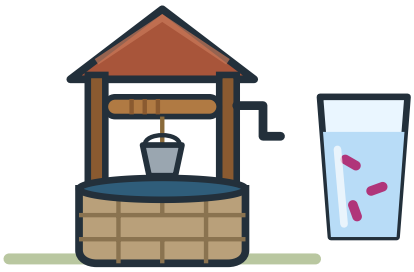
HOW OFTEN IT IS FOUND: Cassia cinnamon is high in coumarin, Ceylon about 100 times lower

HOW OFTEN IT IS EATEN: More at the holidays; some daily in porridge

WHAT WE DO SET 2 72

Well water

E. coli · enterococci · nitrate



WHAT IT DOES

Bacteria cause diarrhoea; nitrate stops a baby's blood carrying oxygen

WHERE IT IS FOUND

Water from your own well for drinking, tea, cooking and baby formula

HOW OFTEN IT IS FOUND: In Latvia, gut bacteria were found in five of six wells tested

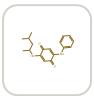
HOW OFTEN IT IS EATEN: Daily in the countryside, rarely in town

RISK ASSESSORS PUT IT HERE

NOT ENOUGH DATA

How well do we know it

?



OTHER

70

6PPD-quinone

6PPD protects tyre rubber from cracking. In air it turns into 6PPD-quinone, which reaches rivers with tyre wear and road runoff. It is deadly to coho salmon. In people it is found in urine, blood and breast milk, but nobody knows how much comes from food rather than dust and air. A 2024 study did not find it in 28 leafy vegetable samples from Switzerland, Italy, Spain and Israel. There is no tolerable daily intake. 'We don't know' is not 'it's safe', and not 'it's dangerous' either.

HOW MUCH IS TOO MUCH?

There is no tolerable daily intake. Neither the source nor a harmful dose for people is known.

Source: Washington State Department of Health (USA), 6PPD and 6PPD-quinone; Sherman et al., 2024, Frontiers in Environmental Science

RISK ASSESSORS PUT IT HERE

MILD × OFTEN

severity × exposure





OTHER

69

Chemicals from packaging and tableware

Packaging, tableware and utensils give a little of their substances to food. This is called migration. The EU sets migration limits, and the national food authority checks goods in shops and at the border. Control works: in 2021 the EU acted against plastic cups made with bamboo powder, which release melamine and formaldehyde. 21 countries found 748 cases, and all the goods were withdrawn. Bisphenol A was different: in 2023 EFSA found that our intake exceeds its new tolerable daily intake, so in 2024 the EU banned it in food contact materials.

HOW MUCH IS TOO MUCH?

Worry is high, but measured exposure is low. Exceedances are very rare, because control finds them and the goods are withdrawn.

Source: European Commission, bamboo and melamine tableware, 2022; Food and Veterinary Service of Latvia (PVD), food contact materials; Regulation (EU) 2024/3190 on bisphenol A; EU Alert and Cooperation Network (ACN) report, 2024

RISK ASSESSORS PUT IT HERE

SERIOUS × SOMETIMES

severity × exposure





WHAT WE DO

72

Well water

The owner is responsible for well water. The Health Inspectorate does not supervise private wells and advises testing them regularly. In a Latvian study (2022) of 356 wells, E. coli was in 9.6 %, enterococci in 82.6 % and too much nitrate in 4.5 %. WHO sets 50 mg of nitrate per litre to protect bottle-fed babies. Boiling kills bacteria but concentrates nitrate. Sometimes covering and cleaning the well is enough.

HOW MUCH IS TOO MUCH?

Have the well tested in a laboratory. If nitrate is high, use other water for baby formula: boiling does not remove it.

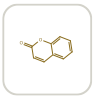
Source: Purmalis and Ozols, 2022, University of Latvia conference; Health Inspectorate of Latvia, 2025; World Health Organization (WHO), nitrate and nitrite in drinking water

RISK ASSESSORS PUT IT HERE

MILD × SOMETIMES

severity × exposure





OTHER

71

Cinnamon and coumarin

Shops mostly sell cassia cinnamon. It holds about 3000 mg of coumarin per kilogram on average; Ceylon cinnamon about 100 times less. BfR and EFSA set a tolerable daily intake of 0.1 mg per kilogram of body weight; an adult reaches it with about 2 g of cassia a day. Since 2011 EU maximum levels apply: 50 mg/kg in traditional and seasonal bakery with cinnamon, 5 mg/kg in desserts. In Austria in 2025, 4 of 151 samples exceeded them.

HOW MUCH IS TOO MUCH?

No need to fear cinnamon. If you eat a lot of it, or every day, choose Ceylon cinnamon.

Source: German Federal Institute for Risk Assessment (BfR), questions and answers on coumarin in cinnamon; Austrian Agency for Health and Food Safety (AGES), 2025; European Food Safety Authority (EFSA); Regulation (EC) No 1334/2008, Annex III

OTHER SET 2 73

Microwaves

Microwaves · non-ionising radiation



WHAT IT DOES

People fear microwaves make food radioactive and “kill” the vitamins

WHERE IT IS FOUND

Microwave ovens at home, at work and in cafés; food heated in them

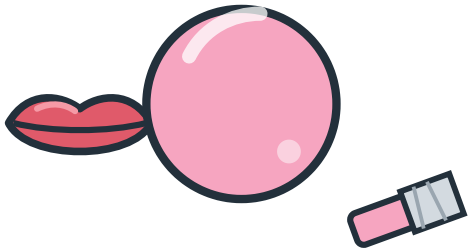
HOW OFTEN IT IS FOUND: The radiation stays in the oven, and in food it turns into heat

HOW OFTEN IT IS EATEN: Every day in many homes

WHAT WE DO SET 2 74

Swallowed chewing gum

Gum base



WHAT IT DOES

They say gum is not digested and stays in the stomach for seven years

WHERE IT IS FOUND

Chewing gum swallowed by accident or out of habit

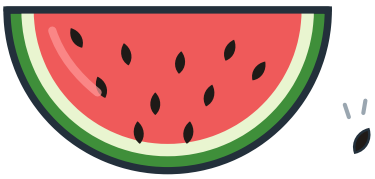
HOW OFTEN IT IS FOUND: Swallowed gum is not digested but passes through the gut

HOW OFTEN IT IS EATEN: Now and then, children especially

WHAT WE DO SET 2 75

Swallowed watermelon seeds

Watermelon seeds



WHAT IT DOES

Children are told a seed sprouts in the stomach and a watermelon grows there

WHERE IT IS FOUND

Watermelon with seeds, eaten in a hurry in summer without spitting them out

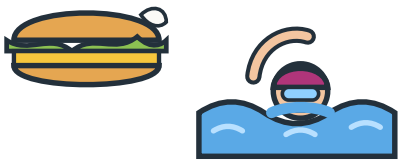
HOW OFTEN IT IS FOUND: A watermelon seed does not sprout in the stomach and passes in about a day

HOW OFTEN IT IS EATEN: Now and then, in summer

WHAT WE DO SET 2 76

Swimming straight after eating

Water · a full stomach



WHAT IT DOES

They say you get cramps in the water and may drown unless you wait an hour

WHERE IT IS FOUND

Lunch on the beach, a picnic meal, an ice cream before a swim

HOW OFTEN IT IS FOUND: Eating before a swim does not raise the drowning risk

HOW OFTEN IT IS EATEN: Now and then in summer: a meal, then a swim

RISK ASSESSORS PUT IT HERE

MILD × SOMETIMES

severity × exposure





WHAT WE DO

74

Swallowed chewing gum

The seven-year story is told to children so they do not swallow gum. It has a true core: the gum base is not digested. That does not mean it stays in the stomach.

A Mayo Clinic gastroenterologist says the story is not true: gum passes through uneventfully and fairly quickly. In 1998 three small children with blockages were described in the US. Two got gum as a reward and swallowed it regularly. The risk is frequent swallowing, not one piece.

HOW MUCH IS TOO MUCH?

One swallowed piece does no harm. Better not to give gum to small children until they can spit it out.

Source: Mayo Clinic; Scientific American on Milov et al., Pediatrics, 1998

RISK ASSESSORS PUT IT HERE

MILD × OFTEN

severity × exposure





OTHER

73

Microwaves

The fear comes from the word 'radiation', which sounds nuclear. Microwaves are non-ionising: they make the water in food hot, and that is all.

The US Food and Drug Administration (FDA) says microwave energy turns into heat and does not make food radioactive. Nutrients are lost no more than in ordinary cooking, and sometimes fewer, because cooking is quick and uses little water. The one tip: the oven heats unevenly, so stir, let the food stand, and keep metal out.

HOW MUCH IS TOO MUCH?

The microwave is safe. Stir the food and let it stand so it heats evenly. Use containers marked microwave-safe.

Source: US Food and Drug Administration (FDA), microwave ovens; Michigan State University Extension, in Food Safety News, 2021

RISK ASSESSORS PUT IT HERE

MILD × SOMETIMES

severity × exposure





WHAT WE DO

76

Swimming straight after eating

The advice to wait an hour is old and sounds logical: digestion needs blood, so the muscles get less. Many people heard it from their parents every summer.

The American Red Cross Scientific Advisory Council reviewed the studies. Eating did not affect swimming, and swimming within an hour of a meal does not raise the drowning risk for children or adults. The real risk lies elsewhere. The US Centers for Disease Control and Prevention (CDC) say alcohol is involved in up to 70 % of deaths of adolescents and adults during water recreation.

HOW MUCH IS TOO MUCH?

Eating before a swim is fine. Alcohol is not, and children in the water need watching.

Source: American Red Cross Scientific Advisory Council, 2024; US Centers for Disease Control and Prevention (CDC), risk factors for drowning

RISK ASSESSORS PUT IT HERE

MILD × SOMETIMES

severity × exposure





WHAT WE DO

75

Swallowed watermelon seeds

The story is told to children so they spit the seeds out. It sounds logical: in soil a seed does sprout, so why not in a belly.

But a seed needs soil with nutrients, and time. Stomach juice is acidic, and the seed stays in the gut for only about a day. Thanks to the hard coat it often comes out almost unchanged. In many parts of the world the seeds are roasted and eaten as a snack. A great many can cause bloating, and a small child can choke on them.

HOW MUCH IS TOO MUCH?

A swallowed seed does no harm. With small children, just watch that they do not choke.

Source: Wonderopolis (National Center for Families Learning); Acibadem Health Point, 2026